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IMPLEMENTATION COMPLETION AND RESULTS REPORT

LN. 8488-IN

ON A

LOAN

IN THE AMOUNT OF US\$400 MILLION

TO

INDIA

FOR THE

TAMIL NADU SUSTAINABLE URBAN DEVELOPMENT PROGRAM

September 22, 2023

Urban, Resilience and Land Global Practice
South Asia Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective August 31, 2023)

Currency Unit = Indian Rupee (INR)

INR 1 = US\$0.0121

US\$1 = INR 82.75

FISCAL YEAR

July 1 – June 30

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ABBREVIATIONS AND ACRONYMS

AMRUT	Atal Mission for Rejuvenation and Urban Transformation
CCP	Chennai City Partnership: Sustainable Urban Services Program
CPF	Country Partnership Framework
CPS	Country Partnership Strategy
CREF	Credit Rating Enhancement Fund
DLI	Disbursement-Linked Indicator
DMA	Directorate of Municipal Administration
DSR	Debt Service Ratio
E&S	Environmental and Social
EIRR	Economic Internal Rate of Return
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
FM	Financial Management
GCC	Greater Chennai Corporation
GIS	Geographic Information System
GoI	Government of India
GoTN	Government of Tamil Nadu
IA	Implementing Agency
ICR	Implementation Completion and Results Report
IPF	Investment Project Financing
IRI	Intermediate Results Indicator
IVA	Independent Verification Agency
JNNURM	Jawaharlal Nehru National Urban Renewal Mission
LED	Light-Emitting Diode
M&E	Monitoring and Evaluation
MIS	Management Information System
MTR	Midterm Review
NRW	Nonrevenue Water
O&M	Operation and Maintenance
OSR	Own Source Revenue
PAD	Project Appraisal Document
PAF	Project-Affected Family
PDO	Project Development Objective
PPP	Public Private Participation
RAP	Resettlement Action Plan
RF	Results Framework
RPF	Resettlement Policy Framework
RTFFS	Real Time Flood Forecast System
SBM	Swachh Bharat (Clean India) Mission
SCADA	Supervisory Control and Data Acquisition System
SEBI	Securities and Exchange Board of India
SWD	Stormwater Drainage
TA	Technical Assistance

TNCRUDP	Tamil Nadu Climate Resilient Urban Development Program
TNHUDB	Tamil Nadu Housing and Urban Development Board
TNSUDP	Tamil Nadu Sustainable Urban Development Project
TNUDF	Tamil Nadu Urban Development Fund
TNUDP	Tamil Nadu Urban Development Project
TNUIFSL	Tamil Nadu Urban Infrastructure Financial Services Limited
TTRO	Tertiary Treatment Reverse Osmosis
ULB	Urban Local Body
UTIS	Urban Tree Information System

TABLE OF CONTENTS

DATA SHEET	1
I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES.....	5
A. CONTEXT AT APPRAISAL	5
B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)	7
II. OUTCOME	9
A. RELEVANCE OF PDOs	9
B. ACHIEVEMENT OF PDOs (EFFICACY)	10
C. EFFICIENCY	15
D. JUSTIFICATION OF OVERALL OUTCOME RATING	16
E. OTHER OUTCOMES AND IMPACTS (IF ANY).....	16
III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME.....	17
A. KEY FACTORS DURING PREPARATION	17
B. KEY FACTORS DURING IMPLEMENTATION	18
IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME ..	19
A. QUALITY OF MONITORING AND EVALUATION (M&E)	19
B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE.....	20
C. BANK PERFORMANCE	21
D. RISK TO DEVELOPMENT OUTCOME	22
V. LESSONS AND RECOMMENDATIONS	22
ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS.....	24
ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION	36
ANNEX 3. PROJECT COST BY COMPONENT	39
ANNEX 4. EFFICIENCY ANALYSIS.....	40
ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS ...	45
ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)	46
ANNEX 7. BENEFICIARY SURVEY	47
ANNEX 8. PHOTOS OF PROJECT RESULTS.....	50



DATA SHEET

BASIC INFORMATION

Product Information

Project ID	Project Name
P150395	Tamil Nadu Sustainable Urban Development Project
Country	Financing Instrument
India	Investment Project Financing
Original EA Category	Revised EA Category
Full Assessment (A)	Full Assessment (A)

Organizations

Borrower	Implementing Agency
India	TNUDF

Project Development Objective (PDO)

Original PDO

To improve urban services delivery in participating ULBs in a financially sustainable manner and to pilot improved urban management practices in selected cities.

**FINANCING**

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
IBRD-84880	400,000,000	352,729,993	352,729,993
Total	400,000,000	352,729,993	352,729,993
Non-World Bank Financing			
Borrower/Recipient	200,000,000	209,000,000	182,370,000
Total	200,000,000	209,000,000	182,370,000
Total Project Cost	600,000,000	561,729,993	535,099,993

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
31-Mar-2015	21-Aug-2015	06-May-2019	31-Mar-2022	31-Mar-2023

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
30-Jun-2020	215.41	Change in Results Framework Change in Components and Cost Reallocation between Disbursement Categories Change in Legal Covenants Other Change(s)
09-Jun-2021	260.72	Change in Components and Cost Reallocation between Disbursement Categories
08-Feb-2022	290.82	Change in Loan Closing Date(s)

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Satisfactory	Satisfactory	Substantial



RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	23-Jun-2015	Satisfactory	Satisfactory	0
02	06-Dec-2015	Satisfactory	Satisfactory	0
03	20-May-2016	Satisfactory	Satisfactory	20.07
04	12-Oct-2016	Satisfactory	Satisfactory	21.57
05	25-Apr-2017	Satisfactory	Satisfactory	38.48
06	02-Dec-2017	Satisfactory	Satisfactory	74.89
07	15-Jun-2018	Satisfactory	Satisfactory	121.59
08	30-Nov-2018	Satisfactory	Satisfactory	148.63
09	25-Jun-2019	Satisfactory	Satisfactory	191.77
10	24-Dec-2019	Satisfactory	Satisfactory	209.37
11	11-Feb-2020	Satisfactory	Moderately Satisfactory	211.49
12	16-Oct-2020	Satisfactory	Moderately Satisfactory	219.00
13	02-Jun-2021	Satisfactory	Moderately Satisfactory	260.72
14	30-Nov-2021	Satisfactory	Moderately Satisfactory	290.82
15	27-May-2022	Satisfactory	Moderately Satisfactory	296.90
16	05-Aug-2022	Satisfactory	Moderately Satisfactory	307.29
17	10-Mar-2023	Satisfactory	Moderately Satisfactory	318.82

SECTORS AND THEMES

Sectors

Major Sector/Sector	(%)
Public Administration	30
Sub-National Government	30



Water, Sanitation and Waste Management		70
Sanitation		30
Waste Management		10
Water Supply		30
Themes		
Major Theme/ Theme (Level 2)/ Theme (Level 3)		(%)
Public Sector Management		30
Public Administration		30
Administrative and Civil Service Reform		9
Municipal Institution Building		21
Urban and Rural Development		70
Urban Development		70
Urban Infrastructure and Service Delivery		30
Municipal Finance		40
Environment and Natural Resource Management		8
Climate change		8
Mitigation		4
Adaptation		4
ADM STAFF		
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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. **Urban context in India.** India's rapid economic growth has been accompanied by an unprecedented urban and spatial transformation. The urbanization rate was around 31 percent (377 million persons) as per census 2011 and was projected to increase to 50 percent by 2030. Urbanization not only benefited urban residents but also helped alleviate poverty in rural areas through migration and stronger rural-urban economic links. However, to accommodate an additional 10 million urban dwellers per year, cities in India faced unprecedented challenges in providing their residents with adequate housing and urban services, addressing slums, creating opportunities for economic growth, and ensuring environmental sustainability. Addressing these challenges would require deepening reform and huge investment in urban infrastructure and services, estimated at US\$870 billion over the next 20 years.¹ Cognizant of the urbanization challenges, the Government of India (GoI) launched several urban development programs, including the '100 Smart Cities', 'Atal Mission for Rejuvenation and Urban Transformation' (AMRUT), and 'Swachh Bharat (Clean India) Mission' (SBM).

2. **Urbanization in Tamil Nadu.** Tamil Nadu is a highly urbanized and industrialized state. As per Census 2011, 48.5 percent of Tamil Nadu's population lived in urban areas, much higher compared to the entire country. The proportion of Tamil Nadu's urban population increased from 24.4 percent in 1951 to 48.5 percent in 2011, making it the most urbanized state in the country. At the time, there were 12 corporations, 124 municipalities, and 528 town panchayats in Tamil Nadu. The state's capital, Chennai, was the sixth most populous city in India and the fourth most populous urban agglomeration. The rapid urbanization led to huge service gaps in the cities in Tamil Nadu; for example, coverage of sewerage and stormwater drains was less than 30 percent, and less than 10 percent of solid waste was treated sanitarily and disposed. City governments also needed to deepen urban finance reforms to generate financial resources, create jobs, and improve urban governance to take on the urbanization challenges. According to the Urban Vision 2023 of the Government of Tamil Nadu (GoTN), an estimated US\$42 billion was required to meet the investment needs of urban local bodies (ULBs) during 2012–2023.

3. **Rationale for World Bank involvement.** The World Bank was well positioned to support the GoTN in its efforts to promote sustainable urbanization, given its long engagement with the GoTN in the urban sector. It supported the establishment of the Tamil Nadu Urban Development Fund (TNUDF) within the Tamil Nadu Urban Infrastructure Financial Services Limited (TNUIFSL) in the late 1990s under the World Bank-financed Tamil Nadu Urban Development Project (TNUDP) II and subsequently financed basic infrastructure and services in ULBs through TNUDF under the TNUDP III during 2005–2014. TNUDF had played an important role in fostering greater access to financial markets for ULBs and had become a model for other states in India. Building on experiences from these earlier engagements as well as international good practices, the Tamil Nadu Sustainable Urban Development Project (TNSUDP) intended to demonstrate innovation in three areas: (a) addressing the gaps in urban services, while focusing on improving environmental sustainability; (b) setting up a credit enhancement facility to increase ULBs' access to the capital market, based on TNUDF's experience in issuing municipal bonds and similar interventions in the United States and elsewhere; and (c) piloting a results-based financing approach to incentivize improved urban governance in selected ULBs, learning from other World Bank projects, such as the Pakistan Punjab Cities Governance Improvement Project.

¹ Report on Indian Urban Infrastructure and Services, High-Powered Expert Committee, 2011.

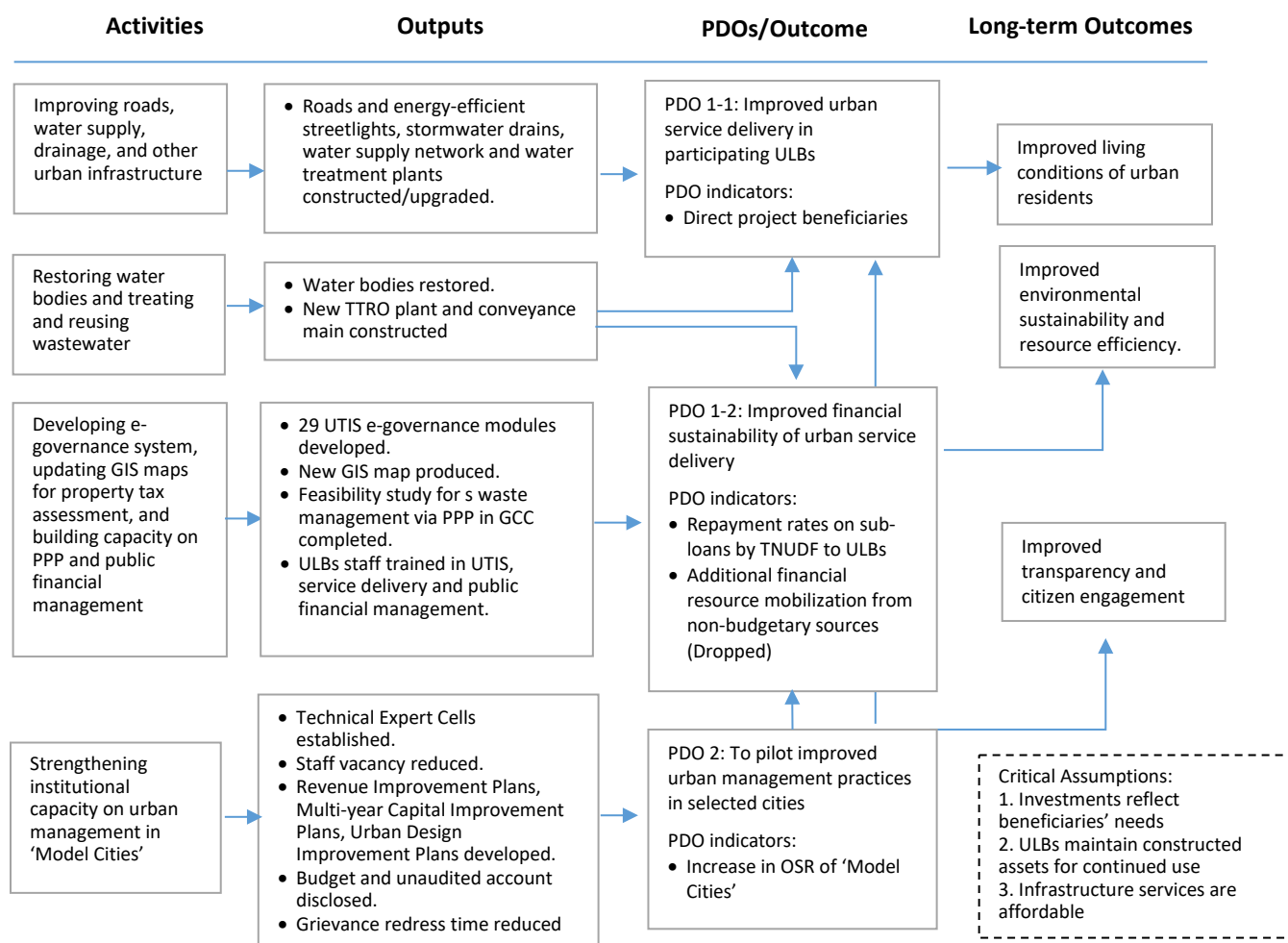


4. Project contribution to higher-level objectives. The project was aligned with the World Bank Group Country Partnership Strategy (CPS) for India for FY2013–17 (Report 76176-IN), particularly Engagement Area 2 on Spatial Transformation that seeks to leverage the rural-urban transformation as an opportunity to reduce poverty and increase competitiveness. It contributed to achieving several outcomes of the CPS, including strengthening the institutional capacity of urban governments, improving urban services, and improving environmental protection. The project was also fully consistent with the urban sector priorities as outlined in India’s Twelfth Five-Year Plan (2012–2017), including increasing investments in urban infrastructure, strengthening urban governance, raising environmental sustainability, and improving the financial sustainability of ULBs. It was also well aligned with GoTN’s Vision Tamil Nadu 2023, which defined the urban vision as “to promote and facilitate the development of inclusive and sustainable cities.”

Theory of Change (Results Chain)

5. The Theory of Change, illustrated in Figure 1, is derived from the project description in the Project Appraisal Document (PAD, Report No. PAD1096).

Figure 1. Theory of Change



Note: PDO = Project Development Objective; PPP = Public Private Participation; OSR = Own Source Revenue; TTRO = Tertiary Treatment Reverse Osmosis; UTIS = Urban Tree Information System.



Project Development Objectives (PDOs)

6. The PDO, as stated in the Financing Agreement and the PAD, was to improve urban services delivery in participating ULBs in a financially sustainable manner and to pilot improved urban management practices in selected cities.

Key Expected Outcomes and Outcome Indicators

7. The PDO consisted of the following two outcomes and key indicators at appraisal:

- For PDO 1 'Improved urban service delivery in participating ULBs in a financially sustainable manner', the key PDO indicators were 'Number of direct project beneficiaries, of which female (%)' (target: 2,000,000; 40 percent) and 'Additional financial resources mobilized from non-budgetary sources in INR crores' (target: 492).
- For PDO 2 'Improved urban management practices in selected cities', the key PDO indicator was 'Increase in Own Source Revenues (OSR) of 'Model Cities' in INR crores over the previous year with the implementation of OSR improvement plans' (target: 22.50).

Components

Component 1: Results Based Grants for Urban Governance (original allocation: US\$61 million, of which IBRD US\$55 million; cost at completion: US\$57.41 million, of which IBRD US\$51.67 million)

8. This component, implemented by the Directorate of Municipal Administration (DMA), provided results-based grants to the three 'Model Cities' on a pilot basis for improving urban management in four areas: (a) ULB empowerment and organizational capacity; (b) spatial development planning; (c) sustainable finances; and (d) e-governance and public disclosure. It also financed support to DMA for project management and consultant services.

Component 2: Investments in Urban Services (original allocation: US\$504 million, of which IBRD US\$323 million; cost at completion: US\$462.81 million, of which IBRD US\$290.51 million)

9. This component, implemented by TNUIFSL, financed: (a) investments in improving urban services, including water, sewerage, municipal solid waste, urban transportation, septage management, and stormwater drainage (SWD), as well as support for project management and supervision; (b) the creation of a Credit Rating Enhancement Fund (CREF) to provide credit enhancement to municipal bonds and other market-based financing instruments that ULBs in Tamil Nadu would use for mobilizing additional resources beyond budgetary sources; and (c) technical assistance (TA) to (i) ULBs to prepare and implement subprojects, including environmental and social (E&S) mitigation actions and PPP arrangements, and to implement credit enhancement measures, (ii) the Greater Chennai Corporation (GCC) for urban flood risk mitigation, and (iii) to TNUIFSL for institutional development.

Component 3: Urban Sector Technical Assistance (original allocation: US\$34 million, of which IBRD US\$21 million; cost at completion: US\$13.88 million, of which IBRD US\$9.55 million)

10. This component, implemented by DMA, financed TA for: (a) developing next-generation municipal e-governance and geographic information systems (GISs) for ULBs; (b) institutional development and capacity building, including training, sector studies, operations and maintenance, and strengthening of public financial management in ULBs; and (c) project management and DMA's incremental operating costs.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

11. The project underwent three Level II restructurings, in June 2020, June 2021, and February 2022.



Revised PDOs and PDO Indicators

12. **There was no revision to the PDO.** The wording and/or targets of the three PDO indicators were revised during the first restructuring in 2020 as follows (annex 1 shows the original and revised indicators):

- PDO Indicator 1: Number of direct project beneficiaries, of which female (%) (target: 2,200,000; 40 percent)
- PDO Indicator 2: Repayment rates on sub-loans by TNUDF to ULBs (%) (target: 95)
- PDO Indicator 3: Increase in OSR of 'Model Cities' in INR crores over the previous year with the implementation of OSR improvement plans (target: Cumulative increase of Rs. 31.93 crore each in Erode and Vellore and increase of Rs. 15.96 crore in Hosur over and above the OSR of 2016-17).

Revised Components

13. **Component 1: Results Based Grants for Urban Governance.** An amount of US\$0.50 million initially allocated for project management and TA to DMA under Component 1 was reallocated in 2020 to Component 2 to provide additional TA support to ULBs.

14. **Component 2: Investments in Urban Services.** In 2020, the CREF was cancelled due to a new regulation issued by the Securities and Exchange Board of India (SEBI) that prohibited trusts such as TNUDF to issue bonds. The amount of US\$20 million initially allocated to the CREF was reallocated to urban investments and TA under Component 2. The total allocation of Component 2 was increased from US\$504 million to US\$516 million, as a result of reallocation of funds from Components 1 and 3 and due to an increase in counterpart funds. In 2021, in the wake of the COVID-19 pandemic, a new subcomponent was added to Component 2 with an allocation of US\$5 million from urban investments to finance oxygen supplies, medical and testing equipment/kits and supplies, essential equipment for case management, essential pharmaceuticals, and training and capacity building related to COVID-19 care.

15. **Component 3: Urban Sector Technical Assistance.** In 2020, US\$3.5 million, mostly savings on account of lower-than-anticipated training expenditures and incremental operating costs, was reallocated from Component 3 to Component 2.

Other Changes

16. **Legal covenant.** In 2020, references to the CREF subcomponent under Component 2 were removed from the legal documents, including the legal covenant relating to credit enhancement.

17. **Intermediate Results Indicators (IRIs).** In 2020, the following changes were made to IRIs: the end target of IRI "Number of participating ULBs with updated GIS maps" was revised from 90 to 30 ULBs; the end target of IRI "Number of participating ULBs with new e-governance systems operational" was increased from 90 to 135; the end target of IRI "Improved ULB empowerment in the 'Model Cities'" was revised from "Each ULB has adopted a sustainability plan for the technical expert cell's functions and responsibilities after the Project's closing date" to "Each ULB has appointed a high quality Technical Expert Cell"; and the end target of IRI "Improved ULB citizen engagement in the 'Model Cities'" was revised from "Each ULB has established an online platform to exchange information with citizens regarding key service delivery parameters" to "Each ULB has implemented all the e-governance modules as identified by CMA."

18. **Disbursement categories.** In 2021, a new disbursement category was added for the COVID-19 expenditures. Loan allocations for the various disbursement categories were also changed in 2020 and 2021 in accordance with the changes to the allocation of funds by component, as discussed in the 'Revised Components' section.

19. **Project extension.** The loan closing date was extended by 12 months from March 31, 2022, to March 31, 2023.



Rationale for Changes and Their Implication on the Original Theory of Change

20. The main reasons for the changes were the following: (a) funding for urban investment subprojects was increased to cover the increased costs of existing subprojects and to add new subprojects to utilize loan savings; (b) the cancellation of the CREF subcomponent due to a change in regulation;² (c) the addition of the COVID-19 pandemic emergency response subcomponent; (d) lowering the target related to GIS mapping given the capacity constraint in DMA and ULBs, and increasing the target related to e-governance system to scale up impacts; and (e) the increase in project duration to complete construction activities that were delayed mainly due to the COVID-19 pandemic, the 2021 state elections, and flooding in Chennai in November 2021. The cancellation of the CREF activity does not affect the PDO outcome in a major way, as the CREF activity represented only 3.3 percent of the overall project cost, and there were other activities under Component 1 and 3 to support increase in ULBs' revenue collection and hence enhance their financial sustainability. Overall, these changes facilitated the achievement of project outcomes because they increased the scope of relevant activities (particularly urban investments, COVID-19 response, and e-governance systems), while removing an activity which was no longer consistent with a change in regulation at the national level.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

Rating: **High**

21. At closing, the PDO is highly relevant to the WBG India Country Partnership Framework (CPF) for FY18–22 discussed by the Board of Directors on September 20, 2018 (Report 126667-IN), particularly Area I: Promoting resource-efficient growth, and the 2018 Systematic Country Diagnostic which recognizes the role of efficient cities as a direct contributor to a resource-efficient growth path for India, as the project supported making ULBs more livable and productive by investing in basic services and by improving urban governance. More broadly, the PDO remains closely aligned with the World Bank's twin goals of eliminating extreme poverty and boosting shared prosperity through extending water supply services to underserved populations, including low-income people in the cities and peri-urban areas.

22. TNSUDP's focus on water supply service improvement, reuse of wastewater, e-governance, and strengthening of the financial and administrative capacity of ULBs is also closely aligned with the objectives of India's flagship missions AMRUT 2.0 and SBM 2.0 (2021–2026), as well as the Tamil Nadu Urban Vision 2031 that aims to achieve universal access to basic infrastructure and services in all ULBs, improve environmental sustainability, and strengthen ULBs' institutional capacity. Given the importance of the urban agenda in Tamil Nadu, at the request of the GoTN, the World Bank is currently supporting implementation of the Chennai City Partnership: Sustainable Urban Services Program (CCP, P175221) and preparation of the Tamil Nadu Climate Resilient Urban Development Program (TNCRUDP, P179189). Both programs build on the achievements of TNSUDP, particularly in terms of scaling up the results-focused approach across sectors (under CCP) and over a broader range of urban management reforms in more ULBs (under TNCRUDP).

² In July 2015, following the approval of TNSUDP, SEBI issued the Issue and Listing of Debt Securities by Municipalities Regulation, 2015, stating that only a municipality or a corporate municipal entity may issue debt securities. As such, neither Water and Sanitation Pooled Fund nor TNUDF, being trusts, could issue bonds. TNUFSL met with SEBI and national ministries to request amendment, but the regulation remained unchanged. TNUFSL explored term loans from local commercial banks, but it was deemed unviable due to the tight domestic credit markets. In view of this, it was agreed during the midterm review (MTR) of the project to cancel this subcomponent.



B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

23. **The efficacy analysis assesses two PDO outcomes**, i.e., improve urban service delivery in participating ULBs in a financially sustainable manner and improve urban management practices in selected cities (namely, the three Model Cities), using the relevant PDO indicators and IRIs (see annex 1.C), as well as other results not measured by the Results Framework (RF). Where applicable, results were compared to non-project areas in the same ULB that had not benefited from major investments in urban infrastructure in the past five years (the control group). All data, unless otherwise stated, came from the Borrower's project completion report and the endline beneficiary survey report (annexes 7 and 8).

24. **A split rating is not applied in this Implementation Completion and Results Report (ICR)** because: (a) the PDO remained unchanged; (b) even though the CREF activity and its associated indicator were dropped, the PDO outcome of 'improve financial sustainability' was captured through other indicators related to loan repayment and OSR; and (c) the project's scope was expanded and targets for the indicators on beneficiaries, OSR³ and e-governance systems were increased.⁴

Outcome 1: Improve urban service delivery in participating ULBs in a financially sustainable manner

Outcome 1.1: Improve urban service delivery in participating ULBs

25. Infrastructure investments under the project focused on constructing and upgrading basic infrastructure in 30 participating ULBs⁵ in Tamil Nadu, including (a) water supply systems, (b) SWD systems, (c) roads and streetlights, (d) restoration of water bodies and improvement of surrounding public spaces, and (e) a Tertiary Treatment Reverse Osmosis (TTRO) plant in Chennai and conveyance lines for wastewater treatment and reuse. Project investments in urban service delivery were identified based on demand from the ULBs, with an emphasis on improving the urban environment. The selection of participating ULBs was based on the existing TNUDF framework which assessed the financial sustainability of ULBs as well as the financial viability of the investments.

26. **Access to improved water sources.** The project installed 1,068.12 km of water supply pipelines (representing an increase of 3.1 times over the baseline), added 79.85 million liters per day (mld) of additional water treatment capacity, and provided 148,411 new household water connections (62 percent out of the 70 percent end target). There has been a drastic reduction in the use of stand posts, public fountains, and water tankers⁶ as the primary source of water post project implementation. Before the project, water supply failed to meet the increasing demand, especially during the dry season. The original water sources also suffered from bad odor, muddiness, and higher content of iron and fluoride, and posed health risks. Households had to use various methods of purification such as boiling, filtering, and using alum/herbs before consuming water. The project provided safe and reliable piped water supply to 774,337 people in 11 ULBs, increasing the coverage of tap water supply from 49 percent before the project to 82 percent in 2022. Per capita supply has achieved the state target of 135 liters per capita per day (lpcd)

³ As per PAD, the target for the PDO indicator on OSR was a cumulative increase of INR 75 crore over year 1 to 6 (annual increase of INR 15 crore for year 3 and 4 and annual increase of INR 22.5 crore for year 5 and 6); as per the 2020 restructuring, the target was a cumulative increase of INR 79.82 crore.

⁴ According to the World Bank Guidance: ICR for Investment Project Financing (IPF) issued on December 9, 2021, para. 60, in the case that "scope of the project expanded. If the project became overall more ambitious, generally a split rating is not applied [...]. Generally, the operation can be assessed on the basis of the more ambitious revised outcomes and outcome targets."

⁵ The 30 ULBs include 14 municipal corporations, 15 municipalities, and one town panchayat. Two municipalities, Pallavaram and Pammal, are now merged with the Tambaram Municipal Corporation.

⁶ The dependency on water tankers has reduced from 18.9 percent in 2015 to 0.8 percent in 2022.



in the ULBs receiving investment post project, i.e., more than twice the baseline supply of 50 lpcd in 2011.⁷ Households now connected to the water supply network have also benefited from a reduction in time taken to collect water and a reduction in working days lost due to waterborne diseases, especially for women who often carried the burden of collecting water from non-pipe sources. This also freed up time for women to engage in home-based businesses or other economic opportunities. About 91 percent of the respondents to the endline survey stated that they were satisfied with the quality of water supplied post project implementation, against 35 percent in 2015. On the contrary, the control group reported an increased dependence on water tankers and community tube wells and an increase in the time taken to collect water due to long queues. Although the project increased water household connections by 1.6 times over the baseline in the project ULBs, it did not achieve the IRI 'proportion of new or rehabilitated water and sewer connections achieved against aggregate target among participating ULBs'. This was due to the substantial delay in implementing the water supply subproject in Nagercoil (which is now near completion),⁸ mainly on account of the COVID-19 pandemic related interruption and the inability of the contractor to mobilize labor and materials, as well as the additional time taken for issuing the revised administrative sanction for the cost overrun. This does not affect much the achievement of outcome given the overall improvement in water supply services described above.

27. Benefits from the TTRO plant. The project also constructed the first TTRO plant in Tamil Nadu, which treated and supplied 30 mld of used water from a wastewater treatment plant via 28.50 km of conveyance main to 11 industries in the Manali area outside Chennai city. Before the project, industrial consumers depended on fresh water from lakes, groundwater, and desalinated water. On project completion, the industrial consumers using the recycled water have reported a 17 percent decrease in manufacturing costs owing to their access to cheaper water from TTRO. Reduced dependency on fresh water for industrial use resulted in more water becoming available for domestic use. The sale of the recycled water since the TTRO commenced operation in 2019 has increased the Greater Chennai Corporation's (GCC's) revenue by INR 192 crore (US\$23.20 million), representing a surplus of INR 30 crore (US\$3.63 million), thus contributing to the city's financial sustainability.

28. Access to improved drainage and reduced flooding. The project built 500.4 km of new SWD network, including 438.8 km in Chennai (151 percent of the target of 290 km), and added 98 kilo liters of additional water storage capacity through the rehabilitation of ponds. Before the project, due to frequent flooding, residents faced issues such as mosquito breeding, difficulties in sending children to school and going to work, health problems linked to standing water, fear of live electric wires and open manholes, and damage to roads. Families incurred increased expenses on health, paying for taxis, purchase of drinking water, and repairs to houses. There has been a huge improvement in the flooding situation after the project, benefiting 1.16 million people in the three ULBs where SWD investments were made. In Chennai and Thoothukudi, flooding hot spots were reduced by 50–70 percent,⁹ and the hours of inundation were reduced from 168 hours to less than half an hour; in Cuddalore, the coverage of SWD increased from 36 to 90 percent, and hours of inundation were reduced from 204 to 12.¹⁰ There was also a reduction in expenditures on house repairs linked to flood damage and a reduction in the number of working days lost. The control group (areas in the vicinity of the SWD investments) also reported spillover benefits (albeit on a smaller

⁷ Twelfth Five-Year Plan by Tamil Nadu State Planning Commission.

⁸ The water supply subproject in Nagercoil is near completion and will start operation in September 2023. Number of targeted water connections in Nagercoil represents 18 percent of the overall project target. If excluding Nagercoil, the IRI would achieve 75 percent.

⁹ For the SWD subproject in Chennai, based on the study conducted by Indian Institute of Technology Madras, during the baseline assessment, 113 hot spots were identified, which post project implementation had reduced to 50; the number of hot spots had also been reduced from 145 to 25 for the missing links subproject. In Thoothukudi, the project covered 70 out of 146 hot spots.

¹⁰ Cuddalore is a coastal city, and all waters drain to the Gedyam river enroute to the sea. Despite increased SWD coverage, the standing time for flood waters is longer because of high water levels in the river due to inflows from all secondary drains that reduces the carrying capacity of the channel.



scale) in terms of reduced spending on house repairs, pumping out water, purchase of drinking water, and health care for waterborne diseases.

29. Access to all-season roads. The project constructed and upgraded 335.88 km of roads. A total of 622,328 people now have access to wider and better-quality roads. Before the project, the roads were in a state of disrepair, since the roads no improvement work had been carried out in the last six to ten years. Natural wear and tear, coupled with the disturbance caused by infrastructure works especially the laying of water supply and sewage pipes, further damaged the roads. The project repaved the roads and constructed SWD with culverts. This resulted in a reduction in the time lost due to congestion and flooding, and an improvement in the walking and bicycling experience. Respondents to the endline survey stated that on average, they saved about 30 minutes in travel time due to reduced congestion, including to access work, schools, and markets.

30. In addition, 17,301 energy-efficient streetlights were provided along the upgraded roads to improve safety and visibility. Before the project, most of the streetlights were conventional fluorescent, mercury, or sodium vapor lights. The project installed higher luminosity energy-efficient light-emitting diode (LED) streetlights that generated savings of 45 percent in energy consumption and monthly savings of INR 1.5 crores (US\$0.18 million) for GCC. The installation of streetlights also enhanced the residents' sense of security, especially for women, the elderly, and children.

31. Improved environment. The project funded the ecological conservation and restoration of several water bodies, including the heavily polluted Periya lake in the Chennai Metropolitan Area. The project constructed peripheral drains to divert sewage to the existing trunk sewers to the Sewage Treatment Plant, cleaned and de-silted the lake to remove floating debris and invasive vegetation, strengthened the lake shore, and built a public square and walking trail using the silt excavated from the lake, as well as reclaimed 5.5 acres through biomining of the legacy waste. This contributed to groundwater recharge, improvement in the aesthetic environment of the area, and the return of the natural fauna and flora.

Outcome 1.2: Improve financial sustainability of participating ULBs

32. Key project activities to improve the financial sustainability of participating ULBs' included: (a) development of the e-governance system UTIS to improve revenue collection; (b) updating of GIS maps to improve property tax assessments; and (c) TA and training to build capacity on public financial management.

33. Increased capacity for debt servicing. TA and capacity support provided under the project contributed to all 30 ULBs being able to service their debts from year-on-year revenues and achieving a 100 percent repayment rate on TNUDF sub-loans (i.e., higher than the target of 95 percent). This is noteworthy, given TNUDF's growing ULB loan portfolio and the increasing complexity of urban investments.

34. Improved revenue collection via e-governance. The project developed and operationalized Urban Tree Information System (UTIS), comprising 29 e-governance modules¹¹, in all 158 ULBs (including 138 municipalities and 20 city corporations) in Tamil Nadu (i.e., exceeding the appraisal target of 90 ULBs by 68 and revised target of 135 ULBs by 23). Before the project, the manual systems in ULBs were characterized by poor financial visibility, leakages in revenues, and increased operational costs. From the citizens' perspective, the absence of an automated system led to a lack of confidence in ULB administration, longer service cycles, mishandling of funds, grievances not being handled, and non-inclusive governance overall. Implementation of the UTIS has contributed to improving the financial sustainability of the ULBs and the citizens' interface with ULBs. ULBs are now able to prepare their annual accounts within two months from the end of the financial year, submit their accounts on time to the Fifteenth

¹¹ UTIS is a web-based workflow application software that integrates 29 modules such as birth and death registration, property tax, water charges, trade licenses, building plan management, accounting system, revenue management, and human resources management.



Finance Commission, and access grants. There are now multiple online and offline options for citizens to pay taxes and service charges, including a one-stop web portal and mobile interface. Post UTIS implementation in 2018, property tax collection efficiency has increased from 36 percent in 2018–19 to 53 percent in 2021–22, leading to an increase of 63 percent, or INR 29,059 crore (US\$3.51 billion) in property tax collection. Notably, by expanding the rollout of the UTIS from 135 to 158 ULBs, TNSUDP contributed directly to an increase of INR 7,728 crore (US\$933.90 million) in property tax collection in 23 additional ULBs, which is much higher than the original target of mobilizing additional INR 492 crore (US\$59.46 million) associated with the CREF. The UTIS also provides real-time information on the ownership and conditions of assets, which enables municipalities to leverage additional resources from the market as required.

35. Increased property tax via GIS mapping. The project financed drone mapping and door-to-door surveys in Chennai to identify and update unassessed and underassessed properties, which generated an additional revenue of INR 65 crore (US\$7.85 million) per year from 14 percent of bills verified. With all the bills verified, GCC's annual tax revenue could increase by about of INR 500 crore (US\$60.42 million). The drone survey has been completed in 14 other ULBs and the mobile application for field survey has been developed; however, digitalization of GIS maps and door-to-door survey in these ULBs could not be completed due to the delay in appointing a GIS specialist in DMA. As a result, the IRI 'Number of participating ULBs with updated GIS maps' fell short of both the original target of 90 and the revised target of 30. These activities are expected to be financed by the follow-on operation TNCRUDP. Incompletion of the GIS mapping in the target ULBs would likely not have an immediate impact on property tax collection, as the field verification of bill variations by revenue staff could only take place after the GIS maps are updated and may take over a year based on GCC experience.

36. Increase in property values as a result of project investments. According to the beneficiary surveys and secondary data from real estate portals, infrastructure improvements under the project have had a positive impact on the property market and are expected to contribute to a further increase in property tax collection by the ULBs. On average, the rentals in the project areas have increased 12 percent in the last five years, which is much higher than the 3 percent increase in the control group. In Kotturpuram and Velachery in Chennai where SWD subprojects were implemented, the rental rates in the last five years have recorded an overall increase of 8 percent and 15.3 percent, respectively. In contrast, in the adjoining control area of Besant Nagar, a decrease of 4.2 percent in the rental rates has been recorded despite it being strategically located with access to public amenities and tourist attractions. Areas surrounding the Periya lake also witnessed a 46 percent increase in rental rates in the last five years, with many new developments springing up.

37. Capacity building on public finance management. The project facilitated the creation of a Public Finance Management Reform Cell in DMA to provide strategic inputs and technical support on implementing and sustaining public finance-related reforms in the ULBs. The cell has strengthened the monitoring system at DMA through developing financial indicators and ranking systems of ULBs and has provided suggestions for improving the accounting records submitted by ULBs in UTIS, including on sharing the reports with the public. The project also provided training on public finance management to the participating ULBs.

Outcome 2: Improve urban management practices in three 'Model Cities'

38. Erode and Vellore municipal corporations and Hosur municipality¹² were selected as pilots ('Model Cities') to receive the annual results-based grants based on their achievement of the targets for four Disbursement-Linked Indicators (DLIs, later called Performance-Based Conditions). This result-based approach was innovative as there was no similar program in the state. These DLIs incentivized improvement in four areas: (a) ULB empowerment and organization capacity; (b) spatial and development planning; (c) sustainable finances; and (d) e-governance, public

¹² Hosur has since been upgraded to municipal corporation.



disclosure, and citizen engagement. These ULBs were representative of medium-size cities with populations between 200,000 and 500,000, and were either newly upgraded or expected to be upgraded from municipalities at the time of project preparation and thus required more capacity support.

39. Improved technical and managerial capacity for urban management. The three Model Cities achieved the targets for the three DLIs related to empowerment and organizational capacity (DLI 1), spatial and development planning (DLI 2), and e-governance, public disclosure, and citizen engagement (DLI 4). They were the first to implement UTIS and experienced its benefits, as discussed above. Based on the Multiyear Capital Investment Plan and the Urban Design Improvement Plan developed with the support of the Technical Expert Cells established in the Model Cities, 11 investment projects were undertaken at a cost of about US\$24 million, by utilizing the grant and OSR. The Model Cities successfully reduced staff vacancies to less than 5 percent of sanctioned posts (i.e., much better than the target of 20 percent); succeeded in resolving about 80 percent of grievances within the time limit set in their Citizen's Charters, thus achieving the target; and achieved the target of 90 percent for the Ministry of Urban Development Service Level Benchmarks Index related to solid waste household collection coverage.

40. Increased OSRs. The Model Cities developed revenue improvement plans and achieved the OSR targets for the first four years. Before 2020, field verifications by the ULBs' revenue staff had contributed to the expansion of the property tax base and led to increases in property tax collections and non-tax items such as rent from markets, community halls, plot regulation charges, and solid waste user charges. While the yearly targets from 2020 were not achieved due to the GoTN's decision to roll back the property tax rate increase in November 2019 and lower tax filing and collection during the COVID-19 pandemic, the OSR increases exceeded the yearly target in Years 3 and 4. As a result, the Model Cities achieved a cumulative OSR increase of INR 109 crore¹³ (US\$13.17 million) in 2021–22 over the OSR of 2016–17 (136 percent of the end target of INR 79.82 crore).

41. Success in piloting the results-based grants. The project released grants of INR 357 crores (US\$43.14 million) to the three Model Cities based on their achievement of DLIs. The ULBs used the grants to expand investments in service delivery and to develop revenue-generating amenities such as commercial complexes, community halls, parks, and gyms. The overall disbursement of this component reached 94 percent of the appraisal amount, indicating success of the pilot.

Justification of Overall Efficacy Rating

Rating: **Substantial**

42. The project substantially achieved and exceeded its expected outcomes as measured by the three revised PDO indicators and their respective end targets. At completion, there were 2.8 million beneficiaries with improved infrastructure services in the participating ULBs (Outcome 1.1), which was 140 percent of the appraisal target of 2 million and 127 percent of the revised target of 2.2 million set in 2020. Of these, 49 percent were women, which was higher than the target of 40 percent. The repayment rate of TNUDF sub-loans to the ULBs was 100 percent (Outcome 1.2), exceeding the target of 95 percent. The three Model Cities achieved a cumulative OSR increase of INR 109 crore (US\$13.17 million) in 2021–22 over the OSR of 2016–17 (Outcome 2), which was 136 percent of the end target of INR 79.82 crore. The project achieved six of the eight IRIs, including three IRIs on Model Cities related to solid waste collection coverage, citizen engagement, and empowerment; two on investments related to the length of SWD investments in Chennai and the proportion of ULBs reporting improved services delivery; and one on e-governance. The IRIs related to GIS mapping and new or rehabilitated water and sewer connections were not achieved for the reasons indicated earlier. The unavoidable cancellation of the CREF activity due to exogenous reasons was compensated by other activities which help achieved the same outcome. Overall, the project generated

¹³ This includes the increase of INR 34.33 crore in Vellore, INR 42.08 crore in Erode, and INR 32.77 crore in Hosur.



positive long-term socioeconomic outcomes for the urban households in the participating ULBs, including improved living conditions and environmental sustainability, as well as improved transparency and citizens' trust in administrative services.

C. EFFICIENCY

Assessment of Efficiency and Rating

Rating: **Substantial**

43. Economic analysis. At appraisal, the project adopted a framework approach for the economic analysis of municipal subprojects under Component 2, whereby the analysis would be prepared and appraised prior to their implementation. However, a cost-benefit analysis was conducted for the SWD subproject in GCC, which estimated the economic internal rate of return (EIRR) at 15.5 percent. At completion, economic analysis was performed to assess the economic viability of the different types of investment subprojects based on their completion costs and estimated benefits. The updated economic analysis confirms that investments in SWD, water supply, road improvements, streetlighting, and TTRO were economically viable with the EIRRs in the range of 15–21 percent, i.e., higher than the discount rate of 12 percent (see annex 4 for details). The EIRR for SWD at completion was 16.40 percent, slightly higher than the appraisal stage estimate of EIRR of 15.5 percent. The analysis also indicated that investments in water supply subprojects were cost-effective: the investment unit cost was about 71 percent of similar projects in Karnataka.¹⁴

44. Financial analysis. At appraisal, financial analysis was carried out based on TNUDF's existing methodology for financial appraisal of subprojects: (a) at the subproject level, it was assumed that water and wastewater subprojects would not be financially viable based on subproject revenues alone; and (b) at the ULB level, an aggregate assessment was carried out for all ULBs to assess whether the subprojects were financially sustainable. This assessment indicated that the overall borrowing capacity of all TN ULBs over the following five years was over US\$600 million; hence, the ULBs would be in a position to absorb TNSUDP sub-loans of about US\$150 million. At completion, the financial analysis confirmed that while tariff collection ranged from 32 to 81 percent in the ULBs where water supply subprojects were implemented, the ULBs are meeting water supply operations and maintenance (O&M) costs through a combination of tariff collections, the portion of property tax earmarked for water supply, and subventions from ULB general funds. At the ULB level,¹⁵ all 21 ULBs analyzed showed a total expenditure to total revenue ratio of less than 1, implying strong revenue flows. In addition, they all had a debt service ratio (DSR) of less than 15 percent,¹⁶ indicating their ability to service their debts through annual revenues (see annex 4 for details). The average monthly flat fee of less than INR 300 (US\$3.63) for water is about 1–2 percent of the average household income in the ULBs and is hence below the affordability threshold of 5 percent normally used by the World Bank.

45. Efficiency of implementation. The project was extended by one year, mainly because of delays in construction works due to the COVID-19 pandemic and the extreme weather events in 2021. During the extended period, additional subprojects including road improvements in 15 ULBs and missing SWD links in Chennai were implemented using loan savings, thereby enhancing project outcomes. The one-year closing date extension is less than the average closing date extension in the current India portfolio (19.6 months). Government provided about 90 percent

¹⁴ The per capita cost of the water supply project in Tamil Nadu was INR 22,000 whereas that of the state of Karnataka under AMRUT was INR 30,781.

¹⁵ Financial analysis was conducted for 21 out of 30 participating ULBs where data were made available.

¹⁶ The DSR is calculated as the 'Total Debt Service' divided by 'Total Revenue excluding Grants', where Total Debt Service is a sum of 'Annual Loan Payments' and 'Interest Payments'. DSR of less than 30 percent (0.3) is considered good.



of the counterpart funds estimated at appraisal and despite the significant depreciation of the INR against the US dollar, 88 percent of the loan was disbursed.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

46. **The overall outcome was assessed as Satisfactory** based on the rating of High for relevance of the PDO and the Substantial ratings for efficacy and efficiency.

E. OTHER OUTCOMES AND IMPACTS (IF ANY)

47. **Gender.** The project ensured equal rights for both men and women to participate in consultations and provided equal opportunities for men and women to access the services provided by the project. About 49 percent of the direct beneficiaries were women; as discussed earlier, women benefitted from the project investments for improved water supply and street lighting. At the endline survey, 68 percent of the female respondents in Chennai said there was an increased sense of safety to commute after sunset owing to the installation of streetlights.

48. **Institutional strengthening.** Institutional strengthening was an integral part of the PDO and Components 1 and 3 as discussed in the Efficacy section, but it also strengthened ULBs' institutional capacity in the following areas:

- (a) **Inter-departmental coordination.** The Real Time Flood Forecast System (RTFFS) for Chennai developed under the project allows the sharing of real-time data and the issuing alerts to various stakeholders based on rainfall predictions three days in advance. Information from the system is being used by the Water Resource Department to monitor lake operations and river flows during rains, by the Revenue Department to aid in flood relief measures, by the Police Department and the Fire Department to arrange flood relief and rescue works in flood-affected areas, and by the Education Department to plan school schedules during flood affected days. There is also a public-facing application to share and crowd-source inundation information from the public. The project financed GIS data center in GCC hosts spatial data and coordinates with other departments in their use.
- (b) **Participatory approach.** The project enhanced the capacities of the ULBs to plan and implement infrastructure investments through a participatory process of engaging with communities.
- (c) **E&S management.** The project built the capacities of ULBs on E&S management and demonstrated that the Environmental and Social Management Framework (ESMF) could be used successfully as a tool for E&S safeguards management. The GoTN has subsequently updated and continued to use the ESMF to meet the safeguard requirements of other externally aided projects.
- (d) **Capacity building of ULBs.** The project delivered over 30 training programs which benefited 44,627 participants across various functions, including engineers, town planners, town planning inspectors, sanitary workers and supervisors, and ULB-level assistants. The main topics covered included UTIS, hydraulic modelling and the design of water supply and sewerage systems, supervisory control and data acquisition system (SCADA), urban planning, public finance management, and project management.

49. **Mobilization of private capital.** The project demonstrated the scope of and opportunities for engaging the private sector in municipal service delivery. The TTRO subproject used a design, build and operate contract with a private partner responsible for O&M for 15 years. The project also financed the preparation of a feasibility study for improving solid waste management in GCC through PPP. The private sector has since been engaged in the remediation and management of two existing dump sites, construction of a waste-to-energy plant, and collection and transportation of wastes.

50. **Poverty reduction and shared prosperity.** Investments under the project have directly benefited the urban poor. Data from a sample of ULBs show that water supply investments in four ULBs have benefited 53,357 urban



poor, and road investments in 11 ULBs have improved the connectivity of more than 30 slums. The project benefited the poor through its contributions to a marked improvement in the quality and duration of water supply, a decline in annual expenses linked to flood damage, as well as reduced expenditures for the purchase of potable drinking water, and lower medical costs from waterborne diseases.

Other Unintended Outcomes and Impacts

51. Climate adaptation and mitigation. Flood risk maps generated from the RTFFS in Chennai provide stakeholders with key information to identify key gaps in underlying infrastructure, explore blue green infrastructure strategies, and guide planning to manage climate risks. During the cyclone 'Mandous' in Chennai in December 2022, state agencies used the project developed GIS to identify the road stretches that would be affected by the uprooting of trees and took precautionary measures to reduce potential damages. The restored Periya lake, along with other small waterbodies which were improved as part of SWD subprojects, serves as passive blue infrastructure to store rainwater. Municipal buildings constructed in Vellore using the Model City grant include rainwater harvesting structures and LED lighting. Energy-efficient streetlighting in Chennai and the solar panels installed for District Metered Area control in Namakkal have reduced the cities' carbon footprint in addition to saving energy. Use of wall-mounted meter boxes instead of ground-level meters helped ease meter reading, while reducing the damage from flooding. The TTRO subproject, through supplying recycled wastewater to water-intensive industries, reduced their dependency on fresh water/groundwater and improved water security for domestic use.

52. Preparedness for future disease outbreaks. The COVID-19 emergency response subcomponent contributed to improving the resilience of the health system to disruptions caused by subsequent COVID-19 epidemics. In addition, project investments in water supply are expected to be instrumental in preventing disease outbreaks and protecting human health during future infectious disease outbreaks with fecal-oral transmission.

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

53. Strategic alignment and government commitment. The project was very well aligned with the Government's national and state priorities at appraisal, which generated strong commitment from the GoTN. The project also engaged the same implementing agencies (IAs) that were involved in earlier World Bank-financed projects and hence were familiar with World Bank policies and procedures. As required by GoI, subprojects equivalent to 30 percent of the project cost were ready for procurement by Board approval. The participating ULBs demonstrated their strong commitment to the project by identifying available government land for service infrastructure.

54. Quality of project design. The project design incorporated lessons from earlier projects in Tamil Nadu and elsewhere. It leveraged the TNUDF which has proven to be an effective mechanism for providing finance to a large number of ULBs. It introduced for the first time in the urban portfolio in India a result-based grant to incentivize urban reforms in ULBs. The design of this activity benefited from an urban finance and management assessment funded by the Quality Infrastructure Investment Trust Fund, as well as lessons from other World Bank projects such as the Pakistan Punjab Cities Governance Improvement Project. The urban investments component considered the urgent needs for SWD improvement in Chennai after the severe flooding in December 2015. To improve ULBs' capacity in O&M, Component 1 provided incentives to the Model Cities to prepare Multiyear Capital Investment Plans that would include an assessment of the O&M needs of various urban services and the identification and mobilization of funding sources. During subproject appraisal, TNUIFSL also supported ULBs in assessing the financial sustainability of water supply services and in setting the appropriate level of water tariff to meet O&M needs, and the recommended actions such as phased increase of tariffs and expected improvement in collection efficiencies



were included as part of TNUISFL's loan/grant conditions for ULBs. The PDO statement was realistic, and the project's interventions supported its achievement. The RF was overall well designed.

55. Risk assessment and mitigation measures. The overall risk was assessed as Substantial, with Substantial risks on sector strategies and policies, technical design, fiduciary, and E&S. The risk rating and mitigation measures appeared to be overall appropriate. However, given the procurement deficiency during implementation (see Procurement section), the risks related to market entry and TNUISFL's capacity in supervising other IAs' procurement activities could have been addressed more strongly. In addition, although the challenges related to credit enhancement being a new activity in the India program were acknowledged, the risk associated with the regulatory and enabling environment could have been assessed more thoroughly.

B. KEY FACTORS DURING IMPLEMENTATION

56. Utilization of loan savings. In 2021, the project utilized US\$32 million in loan savings¹⁷ to add additional investments in road improvement in 15 ULBs and for the construction of missing links in the Chennai SWD system. The additional activities already had Detailed Project Reports, did not involve land acquisition or resettlement, and did not incur substantial E&S risks. Timely utilization of the savings provided the GoTN the much-needed funds given the fiscal crisis it was facing due to the COVID-19 pandemic and allowed an expansion of the project's scope within the one-year extension. However, due to the steep depreciation of the Indian rupee against the US dollar,¹⁸ especially during the last year of project, which resulted in a significant increase in the loan amount in Indian rupees, as well as cost savings there was an uncommitted balance of US\$47.27 million (11.8 percent of the loan) as of July 31, 2023, which has been cancelled.

57. Implementation delays. Several factors resulted in implementation delays during 2020–2021: (a) lockdowns and restrictions during the COVID-19 pandemic resulted in the stoppage of works and diversion of ULB staff working on the project, leading to an overall delay of about 9–10 months; (b) the 2021 elections in TN, which resulted in a new government in TN, as well as flooding in Chennai in November 2021; (c) internal government approvals taking much longer than anticipated, while clearances were also required from multiple line departments for the water supply lines to pass through the railways; and (d) changes in design resulted in cost escalations and required further approvals. Some water supply subprojects had to change the siting of water storage structures because of protests from the public, inappropriate site characteristics (such as soil strata), or the presence of railway lines, resulting in significant delays and cost escalations.

58. Stakeholder participation. Consultations with various stakeholders, including the beneficiaries, at each stage of the project ensured that the perspectives of local communities were considered during project design and implementation. In water supply subprojects, stakeholders were involved in identifying the sites for overhead and underground tanks in ULBs where government lands were not available. Public consultations also aided in identifying road stretches and inundated streets most in need of improvement and for identifying areas for urban design improvements to utilize the grants received by the Model Cities. Based on the strong views of residents and shopkeepers located along the Chennai SWD canals to move to the new resettlement site, GCC reduced the width from the earlier 13 meters to 10 meters, thereby avoiding impacts to existing structures and reducing the resettlement impacts from the originally anticipated 330 families to 91 families.

¹⁷ These savings were mainly due to (a) removal of the Credit Enhancement subcomponent, (b) dropping of the Hosur sewerage subproject, and (c) exchange rate savings.

¹⁸ The US dollar to Indian rupee exchange rate was 62.56 at appraisal and 82.18 at closing.



IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

59. The project's Theory of Change established clear links between project activities, results, and the PDO. The RF presented measurable indicators with clearly defined baselines and targets, which were in line with the M&E capacity of the client and the expected availability of data. The RF also measured the number of beneficiaries in a gender-disaggregated manner. However, one PDO indicator had to be dropped due to the cancellation of the related activity and was replaced by another indicator that was originally designed as an IRI. Targets of two PDO indicators and one IRI were also revised during the restructuring in 2020.

60. The M&E system was robust, with roles and responsibilities for M&E clearly defined among the different IAs, including TNUIFSL, DMA, and ULBs. TNUIFSL developed a management information system (MIS) for monitoring physical and financial progress, as well as compliance of various subprojects under Component 2 with the ESMF. The DMA also put in place a web-based MIS through UTIS for activities under Components 1 and 3. The two MIS' integrated the functions of investment planning, project progress monitoring, contract management, withdrawal applications and payments, capital budgeting, and financial management (FM). An independent verification agency (IVA) was mobilized before negotiations to verify results on the DLIs of the Model Cities component. Information collected by ULBs and the IVA was to be compiled and verified by TNUIFSL and DMA, respectively, and consolidated by TNUIFSL and reported to the World Bank on a quarterly basis. The methodology and instruments for the endline beneficiary survey were to be developed in November 2022 to allow sufficient time for implementation before project closing.

M&E Implementation

61. Quarterly reports were generated by TNUIFSL using the two MIS' to assess the physical, procurement, and financial progress in each ULB and identify outstanding issues. There were dedicated staff in each ULB, the DMA, and TNUIFSL in charge of M&E to monitor and collect data on indicators. The quality and timeliness of progress reports submitted to the World Bank were generally good. TNUIFSL, the DMA, and their regional offices carried out joint on-site supervision regularly (often monthly) to the participating ULBs. TNUIFSL provided thorough review reports on project progress, with information on physical and financial progress and the status of achieving the PDO-level indicators and IRIs. These contributed significantly to identifying the main issues for discussion during the MTR and the subsequent restructuring in 2020. Independent verification of the Model Cities component DLIs was carried out by the IVA and verified by TNUISFL to enable the timely release and utilization of the results-based grants. The comprehensive beneficiary endline survey was completed in March 2023. These data allowed a meaningful assessment of the project outcomes and their attribution for the ICR.

M&E Utilization

62. The quarterly M&E reports were used by the World Bank, TNUIFSL, and the DMA to track implementation progress on an ongoing basis and identify issues and potential risks, particularly those related to loan savings, delays in site clearance, and any contract management issues. These reports and the midterm report directly informed the three restructurings. TNUIFSL used the M&E reports to update the leadership in the TN Municipal Administration and Water Supply Department on monthly progress and critical issues for their coordination and support.

Justification of Overall Rating of Quality of M&E

63. **The overall quality of the project M&E was Substantial** given the comprehensiveness of the M&E system's design; the efforts taken by the TNUIFSL, the DMA, and the ULBs to collect, evaluate, and report on progress on



time and with high quality; and the usefulness of the M&E reports in the identification of implementation issues and their resolution. The RF was timely updated in the 2020 restructuring; however, targets of a few IRIs, e.g., the one on GIS mapping, could have been recalibrated during the restructuring in 2021 and/or 2022.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

64. E&S safeguards. The project was classified as Environmental Assessment Category A (Full Assessment) and triggered two safeguard policies: Environmental Assessment (OP/BP 4.01) and Involuntary Resettlement (OP/BP 4.12). During project preparation, the borrower prepared an ESMF, which was reviewed, approved, and disclosed in-country in March 2015 before project appraisal and in the World Bank InfoShop. For the COVID-19 pandemic-related restructuring, the ESMF was updated and redisclosed both in-country in September 2022 and on the World Bank's external site. At closing, the rating for environmental safeguards was Satisfactory, while it was Moderately Satisfactory for social safeguards.

65. Environmental safeguards policies were complied with. The ESMF provided a framework for screening, assessing, and identifying the potential impacts of subprojects; carrying out appropriate environmental assessments; and integrating mitigation measures in compliance with the operational safeguard policies of the World Bank. All subprojects were screened to identify the risks and impacts based on their locations and the types of activities. Safeguard documents, including Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs), were prepared and implemented in line with the ESMF to manage the E&S risks of the investment subprojects and TA activities. In addition to updating progress on environmental safeguards management in the Quarterly Progress Reports sent to the World Bank, independent annual E&S audits helped identify the gaps in safeguards management and paved the way for corrective actions. The project also adopted COVID-19 pandemic protocols for works and activities under execution, including sensitization and training of project staff, contractors, and workers on appropriate behaviors in relation to the COVID-19 pandemic.

66. Social safeguards policies were complied with. The SWD subproject in Chennai involved involuntary resettlement of 91 project-affected families (PAFs) in 2017. The project's Resettlement Policy Framework (RPF) was followed, and a comprehensive Resettlement Action Plan (RAP) was prepared and approved by the World Bank. GCC followed the RAP and resettled the 91 Project Affected Families (PAFs) in two resettlement sites developed by the Tamil Nadu Housing and Urban Development Board (TNHADB). The resettlement process was handled satisfactorily by GCC with regular consultations with the PAFs and intensive handholding support to help them integrate with the host communities. However, according to a policy of the GoTN, the PAFs would not be provided the title deeds of their tenements until five years after the resettlement. Owing to the collective efforts of the GoTN, GCC, TNHADB, TNUIFSL, and the World Bank, all but one of the ten PAFs resettled in the Perumbakkam resettlement site had received their title deeds by September 2022. In the Gudapakkam resettlement site, 76 of the 81 PAFs had received title deeds by June 2023, despite some complex land alienation issues. TNHADB has also made available title deeds for the remaining PAFs to come forward and sign¹⁹, and the World Bank continues to monitor this matter closely until it is resolved. A few other subprojects had temporary adverse livelihood or economic impacts. These were managed by the ULBs and TNUIFSL using the ESMF and RPF, as well as subproject-specific ESIAs and ESMPs.

67. A Grievance Redress Mechanism (GRM) was established in each ULB as planned and was accessible to PAFs through multiple channels including phones, websites, and ULB offices. Under the Chennai SWD subproject, petitions were received and duly responded to in October 2016. Additional complaints were received in March 2018 regarding compensation for partially affected structures; subsequently, these were settled, and compensation was

¹⁹ Issuance of title deeds are pending for the remaining five families due to legal heirs not being available on account of deaths in these families and they can come forward to take the title deeds from TNHADB whenever they choose to do so.



paid. ULBs have reported that many complaints were related to the quality of construction and repairs of streets, which were made directly to the contractors and resolved at the contractors' level.

68. The ESMPs were designed and implemented to ensure timely action and compensation for two 'severe' incidents, whereby prompt remedial actions and compensations were ensured for the families of the victims. All incidents were investigated in line with World Bank procedures and registered in the Fatality Tracking System of the World Bank. All incidents have been closed except one that occurred in January 2023, where compensation was offered by the contractor to the victim's family, but the victim's family has sought judicial resolution for determining the compensation amount; this process is currently underway and will be settled as per relevant procedures.

69. **Procurement.** The World Bank procurement policies and procedures were followed in most transactions. Most of the contracts under the urban investment component were awarded early on in the project, and TNUIFSL managed consultancies efficiently. However, procurements of many packages encountered delays, partly due to the lockdowns and disruptions during the COVID-19 pandemic. In almost all the investment packages, there was little or no participation by larger national/international construction firms, indicating possible barriers to entry. Procurement post reviews discovered many red flags, pointing to the lack of effective supervision being provided by TNUIFSL over procurement by other government agencies. The use of e-procurement, which was a major risk mitigation measure, was also delayed significantly. Most of the contracts under the urban TA component were awarded outside of the World Bank's Systematic Tracking of Exchanges in Procurement tool and accordingly could not be covered in the procurement post review by the World Bank. At closing, the procurement rating was Moderately Satisfactory.

70. **FM.** The World Bank FM policies and procedures were complied with. Unaudited interim financial reports and audit reports with unqualified (clean) opinions were submitted on time before the COVID-19 pandemic, but their submission was delayed during the last two years due to delays in getting data from the ULBs and in completion of the audits. Adequate funds were released by the GoTN to the project mostly on time, except in the initial two years due to delays in administrative sanctions. Accounting records were kept up to date. FM staffing in DMA was not adequate, and qualified staff were not available during a major part of the project. In TNUIFSL, due to its growing portfolio, the existing staff handled the FM aspects of the project on a part-time basis. At closing, the FM rating was Moderately Satisfactory.

C. BANK PERFORMANCE

Quality at Entry

71. The World Bank task team processed the project from the Project Concept Note/identification to Board approval in 10 months, which was relatively short compared to the average processing time of 18 months for projects in India. The task team composition was appropriate for the design of the project, and included operational and technical specialists with experience in urban development projects in India (see annex 2). The task team ensure that the objectives of the project were closely aligned with the national agenda and the World Bank's India CPS. It incorporated lessons from other projects in India as well as from other countries in the project design. The task team conducted comprehensive institutional assessments of the ULBs on urban services, governance, and finance; the findings of these assessments informed the technical designs. The project allocated sufficient funding for TA activities, training, capacity building, and implementation support to the IAs and delivered trainings during project preparation on World Bank policies and requirements to address identified capacity risks. The M&E framework was effective, as discussed in the M&E section. However, the broader regulatory environment on municipal finances could have been more closely examined prior to the inclusion of the credit enhancement subcomponent in the project. The design of the GIS mapping activity overestimated the technical capacity of the DMA and the ULBs to carry out the work.



Quality of Supervision

72. The World Bank provided implementation support to the project through 12 full implementation support missions, complemented by many short technical missions. The Bank teams produced candid and comprehensive mission reports and Implementation Status and Results Reports. The task team leader and most team members were based in New Delhi and were able to interact with the IAs frequently, as required. The World Bank team's composition was appropriate to effectively support the project, and additional expertise was mobilized to advise the team during the MTR in 2019. Fiduciary, environmental, and social safeguards specialists participated in the missions and carried out additional just-in-time technical missions as needed to help address identified fiduciary/safeguard weaknesses. The World Bank team organized hands-on training that significantly strengthened IAs' technical and operational capacities.

73. The Bank processed the second restructuring in April 2021 timely and reallocated funding to support the GoTN's response to the COVID-19 pandemic. The project management rating was downgraded to Moderately Unsatisfactory in May 2021, given the persistent delays in several urban investment subprojects and GIS mapping. The Bank team monitored project the progress closely through monthly meetings with the project agencies and timely processed the restructuring to extend the loan closing date by one year.

Justification of Overall Rating of Bank Performance

74. Based on the discussion above on the Bank's contributions to Quality at Entry and Quality of Supervision, **Bank Performance is rated Satisfactory.**

D. RISK TO DEVELOPMENT OUTCOME

75. **Natural disaster and climate change.** Natural disasters, such as flooding and drought, could have a detrimental impact on project assets and the reliability of urban services. The project financed RTFFS and GIS mapping which increase ULBs' capacity to plan and respond to disaster risks.

76. **O&M.** Another potential risk is the sustainability of the institutional capacity involved in the O&M of project assets. ULBs that have improved their water supply systems under the project have either raised or were committed to raising the water tariff to a level sufficient to cover the O&M costs and debt servicing, as recommended by TNUISL during subproject appraisal. However, due to public resistance, tariff increases are often delayed and collection efficiency remains suboptimal. ULBs are putting in place various measures such as door-to-door communications by field staff and raising public awareness to increase water tariff collection and imposing penalties for late payment. ULBs could also explore signing service contracts with third parties for O&M of water supply systems and link payments to service standards. For infrastructure assets that do not generate revenues, an O&M plan should be developed by each ULB which can be linked to its asset management system for planning and implementing maintenance.

V. LESSONS AND RECOMMENDATIONS

77. **Political commitment, policy reform, and provision of incentives are important ingredients for improving the financial sustainability of water services.** Water tariff in Tamil Nadu, like in the rest of India, remains low and is not based on consumption. To address the difficulty in raising water tariffs and insufficient collection, ULBs should be provided incentives to approve bylaws that allow for volumetric billing, with due considerations for low-income customers. Metering can be rolled out on a pilot basis in selected areas with higher levels of services and higher water consumption, with proper systems in place to monitor consumers' usage and address complaints. In addition, data can also provide important evidence to support ULBs' decision-making to address public misconception on



volumetric billing. Data from SCADA in Namakkal show that the level of water consumption is in fact similar between households covered by the 24/7 pilot and households receiving intermittent supply.²⁰

78. Information, education, and communication activities are essential to instill behavioral changes and sustain service provision at the desired levels. Given the challenges related to water tariff collection, the project could have helped ULBs design and implement a more comprehensive communication strategy throughout the entire project duration to continuously engage local communities, raise awareness, and enhance ownership. Consumers should be provided a reliable schedule of water supply and be educated on benefits of volumetric billing in reducing water wastage. Overall, local communities' participation in O&M of municipal assets could be further explored, including mechanisms for people to contribute in terms of time, labor, and/or funding.

79. The municipal development fund model proved to be effective in mobilizing much-needed urban financing and can be further leveraged to access the private market. TNUDF has supported over US\$3.2 billion in urban investments, while maintaining an outstanding track record of 100 percent loan repayments from ULBs. Future projects can support TNUDF to resume its pooled municipal bond program to increase ULBs' access to financial markets, with the funds raised giving priority to green projects such as sustainable water management, climate change adaptation, and energy efficiency.

80. Result-based performance grants should be scaled up to promote urban governance reform in ULBs. The pilot Model Cities component has demonstrated the effectiveness of linking fund allocation to support improved urban governance and citizen engagement, strengthening staffing and technical expertise, mobilization of own resources, and enhancing accountability in delivering services to city residents. The design of future results-based grants can learn from the lessons from this successful pilot under TNSUDP, as well as the GoI flagship programs such as Jawaharlal Nehru National Urban Renewal Mission (JNNURM)²¹ and AMRUT²², to strengthen the link between the sustainability of service delivery and broader ULB-level institutional strengthening and tailor the reform areas and targets to the performance and needs of cities.

81. Institutional strengthening and capacity building of ULBs require sustained support. ULBs have benefited from the various TA, training, and capacity support activities under the project. However, dedicated funding, staffing, and collaboration with training institutions are required to continue developing and delivering structured capacity-building programs to strengthen institutional capacities of ULBs. Building on the momentum of the project, continuous capacity support is required for: (a) updating and improving UTIS and making information available to the public periodically on budgeting, planning, and investments; (b) supporting ULBs in mobilizing resources, building asset registers, strengthening their FM systems, and exploring opportunities for engaging the private sector in improving service delivery, learning from the successful story of TTRO; (c) developing standard operating procedures for the RTFFS, GIS mapping, and property assessment, as well as assigning dedicated teams for inter-departmental coordination and information sharing; and (d) carrying out refresher trainings on topics such as O&M, performance-based contracting, customer management, consumer grievance redressal, meter reading, billing, collection, and climate resilience. The TNCRUDP Program for Results under preparation includes a component focusing on TA and capacity support to the state and ULBs on many of these aspects.

²⁰ The 24/7 pilot in Namakkal under the project covers 250 households. SCADA shows that each household consumes on average 380 liters per day, which is comparable to the 300–400 liters daily consumption by other households receiving daily water supply of three hours.

²¹ The JNNURM was launched in 2005 as a reform-linked grant program. An extensive reform agenda was stipulated, and grant support to states and cities were conditional on progress on service delivery and legislative, institutional, and financial sustainability reforms. However, the reform agenda treated all sizes of cities the same, and many of the intended reforms remained unfulfilled.

²² Under AMRUT 2, 40 percent of grants for infrastructure is linked to the number of household connections achieved. However, the reform for institutional and financial sustainability continues to remain as a separate window and is not linked to grants. Thus, cities were able to draw down funds for infrastructure creation even though their progress on critical reforms remained weak.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: Improve urban services in participating ULBs in a financially sustainable manner

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries	Number	0.00 01-Apr-2015	2,000,000.00 06-Mar-2015	2,200,000.00 30-Jun-2020	2,800,000.00 31-Mar-2023
Female beneficiaries	Percentage	0.00	40.00	40.00	49.00

Comments (achievements against targets):

The result has been achieved by 127.3% compared to the final target. This is an original indicator of PAD, but its end target was revised in 2020. The result is in line with the PDO to improve urban service delivery in participating ULBs. This was achieved by financing new and upgrading of basic infrastructure in 30 ULBs, including (a) water supply systems; (b) stormwater drainage systems; (c) roads and streetlights; (d) restoration of water bodies and improvement of surrounding public space; and (e) TTRO and conveyance lines for wastewater treatment and reuse.



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Repayment rates on sub-loans by TNUDF to ULBs	Percentage	100.00	95.00	95.00	100.00
		01-Apr-2015	06-Mar-2015	30-Jun-2020	31-Mar-2023

Comments (achievements against targets):

The result has been achieved by 105.3% compared to the final target. This was initially an Intermediate Result Indicator of PAD and was changed to a PDO indicator in 2020. The result is in line with the PDO to improve financial sustainability of participating ULBs. This was achieved by financing e-governance system to increase revenue collection, GIS mapping to improve property tax base, and strengthening financial management capacities of ULBs.

Objective/Outcome: Improve urban management practices in selected cities

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Increase in Own Source Revenues (OSR) of 'Model Cities' in INR Crores over previous year with implementation of OSR improvement plans.	Text	INR72 crores (FY13)	INR22.5 crores	Erode and Vellore Corporations have to show increase of Rs. 31.93 crore each in the end of project over and above the OSR of 2016-17 and Hosur Municipality has to show cumulative increase of OSR amounting to Rs. 15.96 crore in the end of Project over and	INR109 crores



		01-Apr-2015	06-Mar-2015	above the OSR of 2016-17 30-Jun-2020	31-Mar-2023
Comments (achievements against targets): The result has been achieved by 136.6% compared to the final target. This is an original indicator of PAD but its end target was revised in 2020. The result is in line with the PDO to improve urban management practices in three 'Model Cities'. This was achieved by financing result-based grants to the ULBs to implement reforms in enhancing empowerment and organizational capacity, spatial and development planning, and e-governance, public disclosure and citizen engagement, and own source revenues.					

A.2 Intermediate Results Indicators

Component: A. Results Based Grants for Urban Governance Component

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Average increase on MoUD Service Level Benchmarks (SLB) Index for the 'Model Cities' related to solid waste household collection coverage.	Percentage	65.00 01-Apr-2015	90.00 06-Mar-2015		90.00 31-Mar-2023
Comments (achievements against targets):					



The result has been achieved by 100% compared to the final target. This is an original indicator of PAD. The result is under the first component (Results Based Grants for Urban Governance) and in line with the PDO to improve urban management practices in three 'Model Cities'. This was achieved by financing result-based grants to the ULBs to implement reforms in urban governance.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Improved ULB citizen engagement in the 'Model Cities'.	Text	Grievances are received offline. Use of online grievance application is limited because of low levels of awareness, technical glitches and poor visibility of the website.	Each ULB has established an online platform to exchange information with citizens regarding key service delivery parameters.	Each ULB has implemented all the e-governance modules as identified by CMA.	All three model cities completed implementation of 29 municipal e-governance modules and also met grievance redressal response time requirement under its citizen charter in 80% of cases
		01-Apr-2015	06-Mar-2015	30-Jun-2020	31-Mar-2023

Comments (achievements against targets):

The result has been achieved by 100% compared to the final target. This is an original indicator of PAD but its end target was revised in 2020. The result is under the first component (Results Based Grants for Urban Governance) and in line with the PDO to improve urban management practices in three 'Model Cities'. This was achieved by financing result-based grants to the ULBs to implement reforms in e-governance, public disclosure and citizen engagement.



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Improved ULB empowerment in the 'Model Cities'.	Text	Authority to approve capital works is low in terms of Administrative Sanction (AS) powers. 01-Apr-2015	Each ULB has adopted a sustainability plan for the technical expert cell's functions and responsibilities after the Project's closing date. 06-Mar-2015	Each ULB has appointed a high quality Technical Expert Cell. 30-Jun-2020	Each ULB maintained a high-quality Technical Expert Cell during implementation. 31-Mar-2023
Comments (achievements against targets): The result has been achieved by 100% compared to the final target. This is an original indicator of PAD but its end target was revised in 2020. The result is under the first component (Results Based Grants for Urban Governance) and in line with the PDO to improve urban management practices in three 'Model Cities'. This was achieved by financing result-based grants to the ULBs to implement reforms in enhancing empowerment and organizational capacity.					

Component: B. Investments in Urban Services Component

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Length of Storm Water Drains constructed / rehabilitated and made operational under the project for Greater Chennai	Kilometers	0.00 01-Apr-2015	290.00 06-Mar-2015		438.80 31-Mar-2023



Comments (achievements against targets):

The result has been achieved by 151.3% compared to the final target. This is an original indicator of PAD. The result is under the second component (Investments in Urban Services) and in line with the PDO to improve urban service delivery in participating ULBs. This was achieved by financing new and upgrading stormwater drainage systems in Chennai.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Proportion of new / rehabilitated water / sewer connections achieved against aggregate target amongst participating sub-projects in these sub-sectors (Percentage)	Percentage	0.00 01-Apr-2015	70.00 06-Mar-2015		62.00 31-Mar-2023

Comments (achievements against targets):

The result has been achieved by 88.6% compared to the final target. This is an original indicator of PAD. The result is under the second component (Investments in Urban Services) and in line with the PDO to improve urban service delivery in participating ULBs. The project financed 148,411 new water household connections, representing 62 percent of the aggregate 240,319 expected connections, lower than the target of 70 percent. This was mainly due to substantial delay in implementing the water supply subproject in Nagercoil, mainly on account of COVID interruption and inability of contractor to mobilize labor and materials, as well as the additional time taken for issuing revised administrative sanction due to cost overrun.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
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Percentage of participating ULBs seeking investment support reporting delivery of improved urban services in their cities.	Percentage	0.00 01-Apr-2015	80.00 06-Mar-2015		96.70 31-Mar-2023
Comments (achievements against targets): The result has been achieved by 120.9% compared to the final target. This is an original indicator of PAD. The result is under the second component (Investments in Urban Services) and in line with the PDO to improve urban service delivery in participating ULBs. This was achieved by financing new and upgrading of basic infrastructure in 30 ULBs, including (a) water supply systems; (b) stormwater drainage systems; (c) roads and streetlights; (d) restoration of water bodies and improvement of surrounding public space; and (e) TTRO and conveyance lines for wastewater treatment and reuse.					
Component: C. Urban Sector TA Component					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
No. of participating ULBs with new e-governance systems operational.	Number	0.00 01-Apr-2015	90.00 06-Mar-2015	135.00 30-Jun-2020	158.00 31-Mar-2023
Comments (achievements against targets): The result has been achieved by 117.0% compared to the final target. This is an original indicator of PAD but its end target was revised in 2020. The result is under the third component (Urban Sector Technical Assistance) and in line with the PDO to improve financial sustainability of participating ULBs. This was achieved by financing e-governance system to increase revenue collection and strengthen financial management capacities of ULBs.					



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of participating ULBs with updated GIS maps.	Number	0.00	90.00	30.00	1.00
		01-Apr-2015	06-Mar-2015	30-Jun-2020	31-Mar-2023
Comments (achievements against targets): The result has been achieved only by 3.3% compared to the final target. This is an original indicator of PAD but its end target was revised in 2020. The result is under the third component (Urban Sector Technical Assistance) and in line with the PDO to improve financial sustainability of participating ULBs. While GIS mapping was completed in Chennai, this activity was not fully implemented in other ULBs. Although drone survey has completed in 14 other ULBs and mobile application for field survey has been developed, digitalization of GIS maps and door-to-door survey could not be completed due to delay in appointing a GIS specialist in DMA and prolonged internal approval process.					



B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1 To improve urban services delivery in participating ULBs in a financially sustainable manner	
Objective/Outcome 1.1 To improve urban services delivery in participating ULBs	
Outcome Indicators	1. Direct project beneficiaries.
Intermediate Results Indicators	1. Length of Storm Water Drains constructed / rehabilitated and made operational under the project for Greater Chennai 2. Proportion of new / rehabilitated water / sewer connections achieved against aggregate target amongst participating sub-projects in these sub-sectors 3. Percentage of participating ULBs seeking investment support reporting delivery of improved urban services in their cities.
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1.1)	Component 2. Investments in Urban Services 1. 148,411 new water household connections 2. 1,068.12 km of new water distribution network 3. 79.84 mld of additional water treatment capacity 4. 98 kl of additional water storage capacity 5. 500.35 km of new SWD network 6. 335.88 km of improved roads 7. 17,301 energy-efficient streetlights installed 8. 508.74 acres of water bodies restored 9. 30 mld of water treated in new TTRO 10. 28.50 km of conveyance main constructed.
Objective/Outcome 1.2 To improve financial sustainability of participating ULBs	
Outcome Indicators	1. Additional financial resource mobilization from non-budgetary sources through credit enhancement and other means under the project (Dropped) 2. Repayment rates on sub-loans by TNUDF to ULBs.
Intermediate Results Indicators	1. No. of participating ULBs with new e-governance systems operational 2. Number of participating ULBs with updated GIS maps.



Key Outputs by Component (linked to the achievement of the Objective/Outcome 1.2)	Component 3. Urban Sector TA 1. 29 UTIS municipal e-governance modules 2. New mobile application for GIS data collection and verification 3. Drone mapping for 14 ULBs 4. New GIS map developed covering 426 km² of GCC 5. New GIS data center set up in GCC 6. RTFFS developed in GCC 7. Feasibility study for solid waste management via PPP in GCC 8. 44,627 staff trained on 31 training programs.
Objective/Outcome 2 To pilot improved urban management practices in selected cities	
Outcome Indicators	1. Increase in Own Source Revenues (OSR) of 'Model Cities' in INR Crores over previous year with implementation of OSR improvement plans.
Intermediate Results Indicators	1. Average increase on MoUD Service Level Benchmarks (SLB) Index for the 'Model Cities' related to solid waste household collection coverage 2. Improved ULB citizen engagement in the 'Model Cities' 3. Improved ULB empowerment in the 'Model Cities'.
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	Component 1. Results Based Grants for Urban Governance 1. Technical Expert Cells established 2. Revenue Improvement Plans developed 3. Multiyear Capital Improvement Plans developed 4. Urban Design Improvement Plans developed.



C. KEY OUTPUTS BY ULBS

S. No.	ULBs	ULB ^a	Population	Area (km ²)	Project Outputs ^b	Project Beneficiaries ^c
1	Avadi	MC	344,701	65	RI: 18.52 km	34,370
2	Coimbatore	MC	1,602,861	246.8	RI: 46.57 km	160,286
3	Cuddalore	MC	1,73,636	101.6	SWD: 32.32 km	61,604
4	Devakottai	M	76,037	12.42	RI: 19.88 km	7,603
5	Erode ^d	MC	498,121	109.5	RI: 29.92 km	49,812
6	Gobichettipalayam	M	59,523	111.11	RI: 23.93 km	5,952
7	GCC	MC	46,46,732	426	SWD: 450.88 km TTRO: 30 mld SL: 17,301	SWD: 1,023,103 TTRO: 220,000
8	Hosur ^d	MC	1,16,821	72.41	—	—
9	Madurai	MC	1,470,754	148	RI: 11.81 km	147,075
10	Nagercoil ^e	MC	2,24,849	61.36	WSI: 41.12 mld WTP and 412.36 km WDN (expected)	280,000 (expected)
11	Namakkal	M	55,145	55.24	WSI: 17.66 mld WTP and 325.66 km WDN RI: 17.30 km	WSI: 133,013
12	Pallavaram ^f	M	2,15,417	18	WSI: 258.00 km WDN WBI: 508.74 acres RI: 12.45 km	WSI: 215,452
13	Pammal ^f	M	75,870	13	WSI: 115.22 km WDN	73,374
14	Paramakudi	M	95,579	13.45	RI: 14.68 km	9,557
15	Pollachi	M	90,124	13.87	RI: 13.50 km	9,012
16	Rasipuram	M	50,244	24	RI: 15.24 km	5,024
17	Puliyangudi	M	66,034	55.16	WSI: 46.08 mld WTP and 284.05 km WDN	57,600
18	Rajapalayam	M	1,30,442	68.01		75,000
19	Sankarankoil	M	57,277	158.2		50,400
20	Sivakasi	MC	71,040	46.27		40,000
21	Thiruthangal	M	55,362	7.9		30,500



S. No.	ULBs	ULB ^a	Population	Area (km ²)	Project Outputs ^b	Project Beneficiaries ^c
22	Thiruvengadam	TP	8,337	15.6		4,575
23	Tambaram	MC	164,830	87.64	RI: 10.45 km	16,483
24	Tenkasi	M	70,545	65	RI: 23.97 km	7,054
25	Theni	M	94,453	148.9	WSI: 16.10 mld WTP and 85.41 km WDN	94,423
26	Thoothukudi	MC	2,37,830	90.66	SWD: 25.75 km	77,193
27	Tiruchengode	M	95,335	124.6	RI: 13.54 km	9,533
28	Tirunelveli	MC	474,832	189.9	RI: 18.86 km	47,483
29	Tiruppur	MC	877,778	159.6	RI: 15.24 km	87,777
30	Vellore ^d	MC	252,039	87.92	RI: 20.55 km	25,203

Source: Primary Census Abstract; Census of India 2011; Borrower's Completion Report.

Note: a. MC - Municipal Corporation, M - Municipality, TP - Town Panchayat;

b. RI - Road Improvement, MC - Model Cities, SL - Streetlighting, WSI - Water Supply Improvement, WBI - Water Body Improvement, WTP - Water Treatment Plant, WDN - Water Distribution Network.

c. Calculation of beneficiaries of water supply is based on actual household connections provided by the project; for TTRO, it is based on population that could be served by the water produced; for SWD it is based on the population located in the catchment of the network; for roads, it is based on 10 percent of population of those ULBs with road investments only, to account for road users living outside of the immediate neighborhoods; for ULBs that had water supply investments besides roads, beneficiaries for roads were considered 0 to avoid double counting; for water body improvement and streetlighting, the beneficiaries were also considered 0 as these were already counted in the SWD and water supply subprojects of the respective ULBs.

d. Erode, Hosur, and Vellore are the three Model Cities. Both Erode and Vellore had road improvement subprojects; Hosur did not have investment projects.

e. WSI in Nagercoil is only expected to be in operation in September 2023; hence, the outputs and beneficiaries are expected numbers and were not considered in this ICR.

f. Pallavaram and Pammal are now merged with the Tambaram Municipal Corporation.

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. TASK TEAM MEMBERS**

Name	Role
Preparation	
Raghu Kesavan, Zhiyu Chen	Task Team Leader(s)
Shanker Lal	Procurement Specialist(s)
Mohan Gopalakrishnan	Financial Management Specialist
Jon Kher Kaw	Team Member
Tarun Shankar	Team Member
Zhiyu Chen	Team Member
Harinath Sesha Appalarajugari	Social Specialist
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Hyoung Gun Wang	Team Member
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Vikram Raghavan	Counsel
Barjor E. Mehta	Team Member
Ahmed A. R. Eiweida	Peer Reviewer
Gopalaswamy Srihari	Social Specialist
Raghava Neti	Team Member
Shashank Ojha	Team Member
Axel E. N. Baeumler	Peer Reviewer
Jyoti Sriram	Team Member
I. U. B. Reddy	Social Specialist



Rakhi Basu	Team Member
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Supervision/ICR	
Raghu Kesavan	Task Team Leader(s)
Shanker Lal	Procurement Specialist(s)
Krishnamurthy Sankaranarayanan	Financial Management Specialist
Zoe Kolovou	Counsel
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Payal Malik Madan	Procurement Team
Mansha Chen	Team Member
Anindo Kumar Chatterjee	Social Specialist
Vedamurthy Angadi Jayadevappa	Team Member
Sivaramakrishnan Kumar	Procurement Team
Deepa Balakrishnan	Environmental Specialist

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY14	11.562	50,511.62
FY15	72.822	521,046.43
FY16	0	17,214.47
Total	84.38	588,772.52
Supervision/ICR		
FY15	0	12,039.81
FY16	49.529	194,329.31



FY17	45.866	238,813.91
FY18	43.630	226,733.84
FY19	39.809	212,175.11
FY20	44.327	208,705.18
FY21	43.628	191,912.99
FY22	31.253	158,661.05
FY23	14.521	83,318.64
FY24	.385	2,214.24
Total	312.95	1,528,904.08

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$ millions)		Revised at 2020 Restructuring (US\$ millions)		Actual at Project Closing (US\$ millions)		Percentage of Approval (%)
	IBRD	Counterpart Fund (CF)	IBRD	CF	IBRD	CF	IBRD + CF
Results Based Grants for Urban Governance Component	55.00	6.00	54.50	6.00	51.67	5.74	94.1
Investments in Urban Services Component	323.00	181.00	325.00	191.00	290.51	172.30	91.8
Urban Sector TA Component	21.00	13.00	19.50	12.00	9.55	4.33	57.8
Front-end Fee	1.00	—	1.00	—	1.00	—	—
<i>Subtotal</i>	400.00	200.00	400.00	209.00	352.73	182.37	—
Total	600.00		609.00		535.10		89.2



ANNEX 4. EFFICIENCY ANALYSIS

Economic Analysis

Introduction

1. At appraisal, the project adopted a framework approach for the economic analysis of municipal subprojects under Component 2, whereby the analysis would be prepared and appraised during project implementation. A cost-benefit analysis was however conducted for the SWD subproject in GCC. The economic benefits were assumed to consist of tangible socioeconomic benefits associated with the reduction in flood damage and intangible benefits associated with improvements in the environment and in the overall living conditions. The EIRR of the SWD subproject in GCC was estimated at 15.5 percent, rendering the investment economically viable. For the urban governance component and urban sector TA component, the economic benefits were assumed to include improved efficiency in the allocation of resources and increased EIRR and sustainability of investments; however, no quantified analysis was conducted.

2. At completion, economic analysis was performed to assess the economic viability of the different types of investment subprojects based on completion costs, number of beneficiaries, and a broader set of estimated benefits.

Methodology and Assumptions

3. A cost-benefit analysis was conducted for investments in water supply, road improvements, streetlighting, SWD, and TTRO, which generate economic benefits and can be measured by outputs (Table 4.1). A cost-effectiveness analysis was conducted for the restoration of water bodies and the investment in RTFFS in Chennai, which generate a wide range of benefits with strong public welfare attributes. Only a 'before-and-after' approach was applied in the cost-benefit analysis, while the measurement of benefits against a 'without-project' counterfactual was not conducted.

4. The economic analysis for the project followed the methodologies of the World Bank and Indian standards and regulations. Prices in costs and benefits were market prices without large deviation from their economic values, so shadow prices and conversion coefficients were not used. Economic costs and benefits were calculated based on the constant price level of 2014.

Economic Benefits

5. The main economic benefits derived from the cost-benefit analysis of the project include:

- (a) **SWD.** Compared to the pre-project scenario, the project area under GCC has shown considerable benefits in terms of avoided damage to properties and infrastructure; avoided health costs; and reduced expenditures on repairs, pumping out water, and purchasing drinking water. The analysis indicated an EIRR of 16.4 percent with only direct beneficiaries considered, which is slightly higher than the EIRR of 15.5 percent at appraisal.
- (b) **Water supply.** Incremental benefits include time savings to fetch water from an alternative source, cost savings to treat fetched water, health benefits from working man-days saved avoided water-related illness, and additional revenue from water charges generated by the government. Safe



drinking water also results in considerable health and human development benefits which cannot be quantified. Replacing tanker trucks with piped water also contributes to a reduction in greenhouse gas emissions. The analysis indicated an EIRR of 16.79 percent.

- (c) **Roads and streetlights.** Incremental benefits include time and cost savings due to the reduced incidence of flooding and congestion. Increase in rental benefits in the project areas due to road improvements was also calculated. Other benefits not quantified in the analysis include improved experience of walking and driving on the streets, increased sense of safety, and reduced carbon footprint due to the usage of LED lights in place of halogen streetlights. The analysis indicated an EIRR of 21.67 percent for roads improvement and 15.84 percent for the installation of streetlights.
- (d) **TTRO.** Incremental benefits include revenue generated by the government agency from water and sewerage charges, meter rents, and annual charges. Industries avoided costs related to individual boring, pumping, and treatment of groundwater or surface water. Given the proximity to the sea, reduced extraction of groundwater by industries also avoided saltwater intrusion in the aquifer, contributing to environmental sustainability. The analysis indicated an EIRR of 20.15 percent.

6. Other economic benefits of the project include:

- (a) **Restoration of the Periya lake.** The project supported the ecological conservation and restoration of the lake, wetlands, and reservoirs with an integrated approach to flood mitigation. A regulation flood wall was constructed to protect the surrounding areas from flooding. The project diverted wastewater from being disposed in the lake and had it treated in the wastewater treatment plant. This also ensured availability of better-quality water for groundwater recharge. There has been a significant increase in commercial land prices and construction of real estate in the area after the project.
- (b) **RTFFS in Chennai.** As per the Asian Development Bank's 2021 report on Integrated Urban Flood Management for the Chennai–Kosasthalaiyar Basin Project (RRP IND 49107-009), the average flood damage cost for the Chennai region would amount to INR 26,700 crore (US\$3.23 billion). The INR 100 crore (US\$12.08 million) investment in RTFFS is therefore considered cost-effective, representing 0.3 percent of the average damage caused by floods.

7. The updated economic analysis confirms that investments in SWD, water supply, road improvement, streetlighting, and TTRO were economically viable with the EIRRs in the range of 15–21 percent being higher than the discount rate of 12 percent.



Table 4.1. Economic Benefits of Selected Investments

Projects	Cost (US\$, millions)	Direct Benefits	Indirect Benefits	EIRR (%)
Water supply subprojects in Namakkal, Pallavaram, Pallam, and Theni	81.23	Direct benefits include time saving to fetch water from an alternative source, cost saving to treat fetched water, health benefits for working man-days saved from water-related sickness, and additional revenue from water charges generated by the government.	Safe drinking water also leads to considerable health and human development benefits. Reduction in disease from clean drinking water leads to children spending more time to attend school and promotes development of human capital. Piped water supply also benefits women who are predominately responsible for water collection. The time saved in water collection will afford them additional time for their children's development and well-being. Replacing tanker trucks with piped water also contributes to reduction in greenhouse gas emission.	16.79
SDW in Chennai	225.67	Direct benefits include avoided damage to properties and infrastructure, avoided health costs and reduced expenditures in repairs, pumping out water, and purchasing drinking water. The estimated saving is INR 751.56 per individual on constructing/repairing properties due to flooding.	-	16.40-16.71
Road improvement subprojects in Tiruchengode, Devakottai, Paramakudi, Pollachi, Gobichettipalayam, Tenkasi, and Erode	13.86	Direct benefits include time and cost saving due to reduced incidence of flooding on a weekly basis and daily congestion. There is also an increase in rental benefits in the project areas due to road improvement.	Other benefits include improved experience of walking and driving on the streets. Female respondents have highlighted the ease of travel leading to more trips on the road. For pedestrians, the ease of walking has reduced the need for taking alternative transport leading to reduced use of fossil fuel.	21.67
Streetlighting in GCC	13.19	Direct benefits include time and cost saving due to reduced congestion and traffic accidents.	Other benefits include improved experience of walking on the streets, increased sense of safety especially for women, and reduced carbon footprint due to usage of LED lights in place of halogen streetlights.	15.84
TTRO	64.00	Direct benefits include revenue generated by the government agency from water and sewage charges, meter rent, annual charges, and so on. Industries can avoid costs related to individual boring, pumping, and treatment of groundwater or surface water.	Given the proximity to the sea, the reduced extraction of groundwater by industries will also avoid saltwater intrusion in the aquifer, contributing to environmental sustainability.	20.15



Financial Analysis

8. At appraisal, financial analysis was discussed based on TNUDF's existing system of financial appraisal of subprojects, which focused on two levels: at the subproject level, most revenue-generating subprojects on water and wastewater were not expected to be financially viable based on subproject revenues alone; at the ULB level, an aggregate assessment of all ULBs was carried out to assess whether the subprojects were financially sustainable for the ULBs as a whole, based on the revenues and expenditures of all ULBs. This assessment indicated that the overall borrowing capacity of all ULBs in Tamil Nadu was over US\$600 million over the following five years; ULBs would therefore be in a position to absorb TNSUDP sub-loans of about US\$150 million.

9. *At completion, at the subproject level*, tariff collection efficiency ranged from 31 to 82 percent in the ULBs where water supply subprojects were implemented (Table 4.2). As a result, ULBs are meeting the O&M costs from other sources, such as ULB general funds and the water charge portion of the property tax.

Table 4.2. Revised Tariff for Water Supply

ULB	Water Tariff - Before (INR per month)		Water Tariff - After (INR per month)		Collection Efficiency (%)
	Residential	Commercial	Residential	Commercial	
Pallavaram	150	750	285	855	39
Pammal	50	100	280	560	33
Theni	50	100	235	705	31
Namakkal	75	150	< 500 sq. ft = 150	< 500 sq. ft = 450	82
			501 to 1,200 sq. ft = 180	501 to 1,200 sq. ft = 540	
			1,201 to 2,400 sq. ft = 210	1,201 to 2,400 sq. ft = 630	
			> 2,400 sq. ft = 270	> 2,400 sq. ft = 810	

10. *At the ULB level*, analysis of 21 ULBs²³ benefiting from the project showed that:

- All 21 ULBs had Total Expenditure to Total Revenue ratios of less than 1,²⁴ implying strong revenue flows (figure 4.1). ULBs need to sustain their OSR by increasing revenue collection efficiency and/or appropriate tariff revisions.
- All ULBs also had a DSR of less than 15 percent,²⁵ indicating their ability to service their debts through annual revenues (figure 4.2).

²³ The analysis was carried out for the 21 ULBs where data were received in March 2023: Avadi, Coimbatore, Erode, GCC, Gobichettipalayam, Madurai, Nagercoil, Pallavaram, Paramakudi, Pollachi, Puliyangudi, Rajapalayam, Rasipuram, Sankarankoil, Sivakasi, Tiruthangal, Tambaram, Tenkasi, Theni, Thoothukudi, and Tiruppur.

²⁴ A Total Expenditure to Total Revenue ratio value of less than 1 is considered good with indication of strong revenue flow; Total Expenditure to Total Revenue ratio higher than 1 indicates deficits in cash flows.

²⁵ DSR of less than 30 percent is considered good.



Figure 4.1. Ratio of Total Expenditure to Total Revenue of Selected ULBs

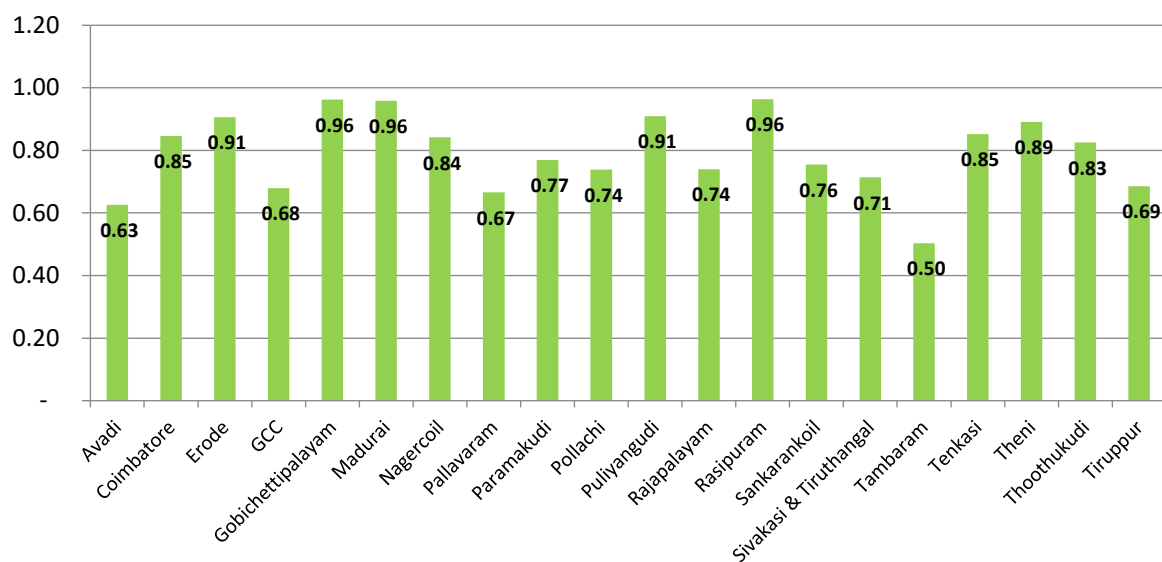
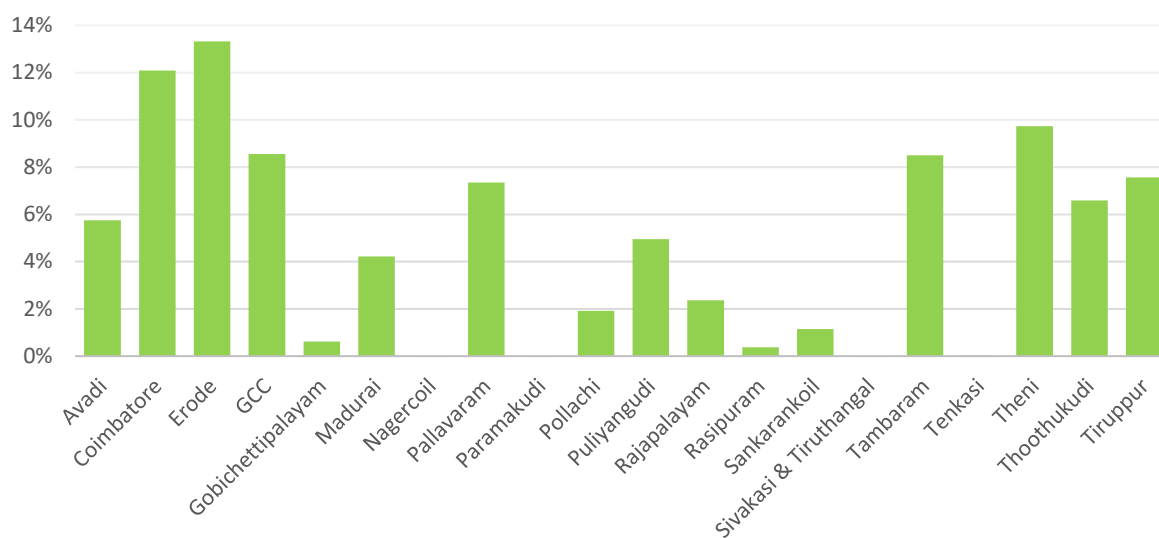


Figure 4.2. DSR of Selected ULBs





ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

1. The World Bank team shared the ICR with the IAs for comments on August 17, 2023, and TNUIFSL responded on behalf of the borrower via email on September 2, 2023 confirming that they have reviewed the ICR and have no comments.



ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)

- Project Appraisal Document (PAD 1096). World Bank, March 2015.
- Loan Agreement and Project Agreement (Loan 8488-IN). World Bank, June 2015.
- Country Partnership Strategy for India 2013–2017 (Report 76176-IN). World Bank, October 2014.
- Country Partnership Framework for India 2018–2022 (Report 126667-IN). World Bank, July 2018.
- Aide Memoires (various) and Management Letters (various). World Bank, 2014–23.
- Implementation Status and Results Reports (various). World Bank, 2015–23.
- Restructuring Papers (Report Nos. RES39021, RES46960, and RES49979). 2020, 2021, and 2022.
- Tamil Nadu Sustainable Urban Development Project Completion Report: Best Practices and Lessons Learnt. Prepared by IPE Global Limited on behalf of TNUIFSL. March 2023.



ANNEX 7. BENEFICIARY SURVEY

1. **Overview.** An endline beneficiary survey was carried out by a consultant hired by TNUIFSL during January–February 2023. The survey included: (a) beneficiary survey and focus group discussions in 17 ULBs covering 23 subprojects (which were completed by December 2022) as well as residents of non-project areas (the control group). In total 7,800 people were surveyed, including 6,240 in project areas and 1,560 in the control group; (b) interviews of 19 state-level and 44 ULB-level staff and consultants involved in project implementation; and (c) interviews of selected project beneficiaries.

2. **Methodology.** The 7,800 sample was generated using multistage stratified systematic random sampling to ensure proportionate geographical representation of the central and outer intervention areas of selected ULBs, as well as socioeconomic representation of selected beneficiaries including female-headed households. Thereafter, a simple random method was applied to select the beneficiaries. The control group was also selected using the same methodology in the areas where investments were not implemented.

3. Summary of Key Outcomes

SWD

- *Reduction in flooding.* There has been a decrease in the frequency and duration of standing flood water. Around 56 percent of the respondents said that they have not observed a single incident of flooding post project implementation. About 54 percent of the respondents had said that they faced floods during the rainy season in 2015, while post implementation, this was reported by only 11 percent.
- *Reduction in working days lost.* About 38 percent of respondents said that they had not lost a single working day due to floods and associated problems post project implementation. Overall, about 70 percent beneficiaries said that there has been a decrease in working days lost after project implementation.
- *Increased household income.* The project led to an increase in the disposable household income of the beneficiaries. About 50 percent of households used to spend money on house repairs and in raising floor levels; this has reduced to 21 percent and 14 percent, respectively, post project implementation. The highest savings were observed in purchasing potable water (87 percent) and medical costs (92 percent).
- *Level of satisfaction.* There is a general high level of satisfaction with respect to the management of construction phase and O&M post project implementation.

Box 7.1. Impact of SWD Investment

A couple living near a low-income settlement by the lake used to struggle to make ends meet due to the waterlogging in front of their house during the monsoon season. The situation was made worse by the mixing of sewage with storm water, which created an unhygienic environment around their home. As a result, their children fell ill with cholera and malaria, leading to unexpected medical expenses and loss of workdays. However, after the implementation of the SWD subproject, their living conditions improved drastically. The waterlogging around their house reduced, and even in low-lying areas, the water gets discharged within 30 minutes. This allowed the couple to resume their daily work without worrying about rain-induced disruptions. The project also brought a sense of hope to all the residents of the area.



Water Supply

- *Increased coverage of municipal water supply service.* There has been a reduction in the use of standposts, public fountains, and water tankers as the primary source of water post project implementation. Around 82 percent of the respondents said they had individual municipal water supply connections compared to 49 percent before the project. Dependency on water tankers has reduced to 0.8 percent in 2022 from 18.9 percent in 2015.
- *Improved quality of drinking water.* About 91 percent of the respondents said that they were satisfied with the quality of water supplied post project implementation; this was against 35 percent in 2015. Before the project, water supply issues included bad odor, muddiness, and higher content of iron and fluoride. Households used various methods to purify water before consumption, e.g., boiling, filtering, and using alum/herbs.
- *Reduction in the time to collect water.* Before the project, 45 percent of the respondents said they spent less than 30 minutes to collect water; this has now reduced to 25 percent. Overall, about 80 percent of the households said that there has been a reduction in the time taken to collect water.
- *Reduction in the working days lost due to waterborne diseases.* About 47 percent of respondents said they had not lost paid working days due to waterborne diseases post project implementation. Before the project, 17.2 percent respondents had said that they had lost up to one week of paid working days, and this has reduced to 7 percent post project implementation.
- *Level of satisfaction.* More than 75 percent of the respondents were extremely satisfied with the water supply scheme implemented, particularly in terms of water pressure, duration of supply, water quality, and regular maintenance of water supply lines.

Box 7.2. Impact of Water Supply Investment

A 33-year-old housewife from Pallavaram recalled the severe water scarcity faced by her family around five years ago. At that time, the family relied on community sources of water supply and had to pay exorbitant sums to water tankers for their daily use, and even then, they sometimes did not get water at all. The family used to spend more than an hour every day in the morning/evening to collect water during the summer and relay water jugs during rainy seasons. In addition, poor water quality led to multiple waterborne diseases. After the implementation of the project, she now has access to clean water via an individual household connection with 1.5–2 hours of supply every alternate day.

Road Improvement and Streetlighting

- *Improvement in the physical conditions of roads.* About 90 percent of the respondents stated that there has been an improvement in the condition of roads.
- *Reduction in time lost due to congestion.* Respondents stated that on average, they used to incur an additional 30-minutes of travel time due to congestion before project implementation.
- *Improvement in the walking, cycling, and driving experience.* About 75 percent of the respondents said that their walking and cycling experience had improved post road improvements and 76 percent of the respondents said that their driving experience had improved. Road improvements also included the development of drains, and this has also contributed to an improvement in managing flood waters. About 32 percent of the respondents said that they had experienced waterlogging in 2015, and this has reduced to 14.6 percent post project implementation.



- *Increased feeling of safety to commute after sunset.* Around 68 percent of the female respondents said that they felt safer to commute after sunset owing to the installation of streetlights in Chennai.
- *Reduction in the time taken for completing the trips.* There had been a reduction in the travel time due to the improvements of the roads. On average, about 19 minutes are saved in work-related trips, 20 minutes in educational trips, 25 minutes in shopping, and 23 minutes in recreation-related trips.
- *Level of satisfaction.* On a scale of 1 to 5, around 80 percent of respondents have given more than 3-star rating in terms of overall satisfaction with the subprojects on road improvement and streetlights.

Box 7.3. Impact of Road Investment

A working woman from Ponni Nagar talked about her safety-related apprehensions while walking along internal residential streets after dusk a few years back. At that time, these streets either did not have streetlights, or the existing lights were not functional. She was especially concerned about incidents of sexual harassment and minor accidents during rains and tried to return home before sunset. She was extremely happy with the streetlights that were installed under the project and feels more confident to travel alone through the streets even after dusk. She observed that minor accidents have reduced, and the streets have become livelier and more women and children are visible on the streets. She rates this as a huge improvement in the 'street experience'.

A woman residing in the same neighborhood was accessible by a muddy and pothole-ridden road, posing challenges to daily mobility. She recalled the increased tension, especially during an emergency. The access conditions worsened after heavy rainfall, and stagnant water made transportation by foot or vehicle even more difficult. The project funded improvement of this last-mile connection to her home, including the construction of drains along the road, drastically improving their accessibility. Families like hers are investing in home improvements knowing that the rains will have a reduced impact.



ANNEX 8. PHOTOS OF PROJECT RESULTS

Component 2: Investments in Urban Services

Road Improvements

Devakottai Municipality

Before



After



Erode Corporation

Before



After



Madurai Corporation

Before



After





Water Supply

Namakkal Municipality (After)



Pammal Municipality (After)



Theni Alinagaram Municipality (After)





Stormwater Drainage

Cuddalore Corporation (After)



Tuticorin

Before



After



GCC

Before



After





TTRO Plant in Chennai

After



Streetlighting in Chennai

After





Restoration of Periya Lake

Before



After

